

A Review on Machine Learning Algorithms to Predict Daylighting inside Buildings

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Abstract

Steep increases in air temperatures and CO₂ emissions have been associated with the global demand for energy. This is coupled with population growth and improved living standards that encourages the reliance on mechanical acclimatization. Lighting energy alone is responsible for a large portion of total energy consumption in office buildings; and the demand for artificial light is expected to grow in the next years. One of sustainable approaches to enhance energy-efficiency is to incorporate daylighting strategies, which entail the controlled use of daylight inside buildings. Daylight simulation is an active area of research that offers accurate estimations, yet requires a complex set of inputs. Even with today's computers, simulations are computationally expensive and time-consuming, hindering to acquire accelerated preliminary approximations in acceptable timeframes, especially for the iterative design alternatives. Alternatively, predictive models that build on machine learning algorithms have granted much interest from the building design community due to their ability to handle such complex non-linear problems, acting as proxies to heavy simulations. This research presents a review on the growing directions that exploit machine learning to rapidly predict daylighting performance inside buildings, putting a particular focus on scopes of prediction, used algorithms, data sources and sizes, besides evaluation metrics. This work should improve architects' decision-making and increase the applicability to predict daylighting. Another implication is to point towards knowledge gaps and missing opportunities in the related research domain, revealing future trends that allow for such innovative approaches to be exploited more commonly in Architectural practice.

Abbreviations:

ANFIS	Adaptive Neuro-Fuzzy Inference System	ISO	International Standards Organization
ANN	Artificial Neural Network	LS-SVM	Least-Squares Support Vector Machine
AR	Auto Regression	MAD	Mean Absolute Deviation
ARE	Average Relative Error	MAE	Mean Absolute Error
ASE _{1000/250h}	Annual Sunlight Exposure	MBE	Mean Biased Error
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers	MLAs	Machine Learning Algorithms
BPNN	Back-Propagation Neural Network	MLR	Multiple Linear Regression
CART	Classification and Regression Tree	MSE	Mean Square Error
CBDM	Climate-Based Daylight Modelling	NB	Naïve Bayes
CDA	Continuous Daylight Autonomy	PC	Predictive Confidence
CFS	Complex Fenestration Systems	PCA	Principal Component Analysis
CGI	CIE Glare Index	PE	Percentage Error
CHAID	Chi-Square Automatic Interaction Detection	R	Coefficient of Correlation
CIE	Commission Internationale de l'Éclairage	R ²	Coefficient of Determination
CNN	Convolutional Neural Networks	RBF	Radial Basis Function
DA	Daylight Autonomy	RBFNN	Radial Basis Function Neural Networks
DC	Daylight Coefficient	RER	Relative Error Rate
DF	Daylight Factor	RF	Random Forest
DGP	Daylight Glare Probability	RMSE	Root Mean Square Error
DT	Decision Tree	RNN	Recurrent Neural Networks
FFNN	Feed-Forward Neural Network	sDA _{300/50%}	Spatial Daylight Autonomy
GA	Genetic Algorithms	SVC	Support Vector Clustering
GP	Gaussian Process	SVM	Support Vector Machine
ID3	Iterative Dichotomiser 3	SVR	Support Vector Regression
IES	Illuminating Engineering Society	UDI	Useful Daylight Illuminance
IESNA	Illuminating Engineering Society of North America		

1. Introduction

Over recent years, steep increases in the air temperatures (Santamouris et al., 2015), which cause extreme weather fluctuations and climate changes (NASA, 2019; Pérez-Lombard et al., 2008), have been associated with the growing demand for energy. In the US and other regions around the world, almost 40% of the total energy is consumed by the building sector (Allouhi et al., 2015; USDOE, 2019), accounting for more than 27% of the related CO₂ emissions (IEA, 2018); while over 75% of the global electricity is generated from non-renewable resources (IEA, 2019a). This is seemingly clear in cooling-dominant climates, where heat gains due to excessive solar radiation is inevitable (Ayoub, 2019a). With the population growth and the improved living standards, the operation of mechanical acclimatization increased as well, consuming more than three-quarters of the total energy to maintain indoor thermal comfort (Koch-Nielsen, 2013). Even with better energy-efficiency practices (IEA, 2019b), building sector continues to demand more energy (De Cian et al., 2007; Santamouris et al., 2015), supported by the expanded capacity to generate electrical power in many developing countries; despite the resulting environmental impacts, such as pollution, global warming, greenhouse effect and ozone depletion (Pérez-Lombard et al., 2008).

On the other hand, lighting energy has accounted for almost 19% of total electricity consumption (IEA, 2006), constituting over 20% of the consumed energy in office buildings (Ürge-Vorsatz et al., 2012). But with the recent technological progress, and the developed daylight standards and regulations to enhance building performance, such as Commission Internationale de l'Éclairage (CIE) (Wyszecki, 1970), International Standards Organization (ISO) (Cianfrani et al., 2009) and Illuminating Engineering Society (IES) (DiLaura et al., 2011), lighting becomes responsible for only 7% of the used energy in buildings (IEA, 2019c). Still, the global demand for artificial light continues to grow by 2.4% annually (IEA, 2006). Moreover, lighting energy represents a main source of heat that has a profound impact on the consumption of cooling energy (Tzempelikos and Athienitis, 2007). Adequate daylighting strategies play an important role in reducing both lighting and cooling energies, especially in harsh climates (Ayoub, 2019a; Iqbal and Al-Homoud, 2007). In fact, a proper consideration of both daylighting and excessive solar penetration can save more than 30% of consumed cooling energy (Wong et al., 2010). Therefore, lighting prediction is imperative to achieve building energy-efficiency (Amasyali and El-Gohary, 2018). One of sustainable approaches to enhance such efficiency (Nasrollahi and Shokri, 2016), while promoting visual and thermal comforts as well, is to incorporate daylighting strategies (IEA, 2000) that entail the controlled use of natural daylight inside buildings (Reinhart, 2014). Such considerations become essential for Architecture and building design. Yet, even with the constant efforts to integrate those techniques into the design process, the used tools and the applied professional methods to predict daylighting performance in buildings are still impractical (Nault et al., 2017).

Early years of daylighting prediction have witnessed several attempts to quantify internal daylighting conditions via simple methods, including diagrams (Millet et al., 1980; Waldram and Waldram, 1923), protractors (Dufton, 1946; Hopkinson et al., 1966), scale models (Aizlewood and Littlefair, 1996), mathematical formulas (Athienitis and Tzempelikos, 2002; Copping, 1987) and rule-of-thumbs (Reinhart

and LoVerso, 2010). Supported by the progress of computer processing power (Koomey et al., 2011; Parkhurst et al., 2006), newly developed methods, known as white-box or engineering, extended such conventional practices to simulation-derived approaches (Ward, 1994) that involve Daylight Coefficient (DC) (Mardaljevic, 2000a; Reinhart and Walkenhorst, 2001) and Climate-Based Daylight Modelling (CBDM) (Mardaljevic, 2000b; Reinhart and Herkel, 2000), which offer more accurate predictions. While daylight simulation methodologies vary in scale and complexity from evaluating simple internal spaces, complex façade fenestrations, to urban contexts, they rely on a common set of steps (Reinhart et al., 2006; Rogers and Goldman, 2006). They involve using physically-based simulation tools to quantify the spatiotemporal luminous conditions at given sensor grid-points within a built environment, accounting for material properties and fluctuations of sky luminance distribution as derived from representative weather data of the study location at designated temporal resolutions (Ayoub, 2019b; Mardaljevic, 2000b; Reinhart and Herkel, 2000). For additional details on different approaches of CBDM, such as 4-Component Method, DAYSIM, 3-, 5-, 4- and 6-Phase Methods, the readers are referred to (Ayoub, 2019b; McNeil, 2014, 2013).

Still, the building design community is confronted by several challenges: (i) Daylight simulations demand a complex set of inputs (Ayoub, 2019b), such as weather data (ASHRAE, 2019; NREL, 1995; NSRD, 2019), sky model (CIE, 2003; Perez et al., 1993), space geometry, location, surrounding context (Ayoub, 2019a), opening configurations (Reinhart et al., 2013) and material properties (Ward, 1992). (ii) Along with potential sources of errors (Labayrade and Fontoynt, 2009; Lim et al., 2010; Ochoa et al., 2012), daylight simulations inherit many difficulties that stem from uncertainties related to complex geometries and material definition that cannot accurately match the reality (Inanici, 2013). (iii) Also, the output of daylight simulations typically yields a huge amount of data that need to be reduced into simplified metrics (Reinhart et al., 2006; Reinhart and Wienold, 2010). Most of daylight metrics, such as Daylight Autonomy (DA) (Reinhart and Walkenhorst, 2001) and Useful Daylight Illuminance (UDI) (Nabil and Mardaljevic, 2006), report the temporal occurrences of visual comfort, but without spatial considerations. Although the approved method of daylight metrics IES LM-83-12 (IES, 2012) expanded such metrics with spatial consideration, and introduced Spatial Daylight Autonomy (sDA_{300/50%}) and Annual Sunlight Exposure (ASE_{1000/250h}), they do not capture both spatial and temporal occurrences all together (Ayoub, 2019b). (iv) Even with today's computers, daylight simulations are time-consuming and computationally expensive. These limitations pushed many researchers to develop more efficient approaches, such as 3- and 5-phase CBDM (Lee et al., 2018; Saxena et al., 2010), along with real-time prediction of building performance (Jones and Reinhart, 2015; McNeil and Lee, 2012; Schardl, 2016). (v) Yet again, architects that lack simulation skills become unexcited to include what they consider complex methods into their practices (Mardaljevic, 2015). Even for experienced practitioners, annual simulations can be tedious to acquire multiple design alternatives in acceptable timeframes. (vi) This bottlenecks the design process of iterative nature, and hinders rapid preliminary approximations during the early design stages (Ibarra and Reinhart, 2009), where many vital decisions are made.

Today, Artificial Intelligence applications are achieving exceptional successes in many vital fields. They are being exploited not only to analyze data, but also in applications of creativity, including image recognition, language processing and others. Worthwhile predictive models that build on Machine Learning Algorithms (MLAs), called *black-box* since the building and its components are unknown, have been and continues to receive much recognition from the building design community due to their ability to handle complex non-linear problems. MLAs act as proxies to heavy simulations, as they learn from relevant data to construct mathematically-fit models. Useful information is then extracted before accurate and rapid predictions can be made from newly input data; and this can be achieved without requiring the original building information or conducting any computational simulations. This domain is one of the most interesting paradigms research community is shifting towards over the past decade. It has long been employed in numerous research activities to predict energy consumption in buildings, paralleled with several review studies. Still, machine learning is underexploited to predict daylighting inside buildings, where architects and researchers are just beginning to leverage from its capabilities.

2. Related Review Studies

Lately, MLAs are being utilized to predict internal daylighting (Kazanasmaz et al., 2009; Lorenz and Jabi, 2017; Yoon et al., 2016; Zhou and Liu, 2015), visual comfort (Chatzikonstantinou and Sariyildiz, 2016; Nasrollahi and Shokri, 2016; Navada et al., 2016), sky and efficacy models (Janjai and Plaon, 2011; Li et al., 2010; López and Gueymard, 2007; Pattanasethanon et al., 2008), weather elements (Chen et al., 2011; Zhang et al., 2017; Zou et al., 2016), shading systems performances (Jain and Garg, 2018), High Dynamic Range Imagery (HDRI) illumination from images (Gardner et al., 2017; Zhang and Lalonde, 2017) or 3D objects (Weber et al., 2018), artificial lighting (Bellocchio et al., 2011; Şahin et al., 2015), among other applications. Moreover, there have been many studies that focused on reviewing the use of MLAs to predict solar radiation (Voyant et al., 2017; Yadav and Chandel, 2014; Zendejboudi et al., 2018), sky model classification (Li and Lou, 2018), weather data (Lou et al., 2017) and building performance (Chakraborty and Elzarka, 2019).

Among recent reviews that considered the prediction of energy consumption is the work of Swan and Ugursal (2009), who identified two approaches to predict energy consumption in buildings, regarding the order of data input: bottom-up that involves statistical and engineering methods, also called white-box; and top-down that is attributed to technological and economic methods. The review compared those approaches in terms of data type and size, input of information, complexity and applications. Zhao and Magoulès (2012) drew four approaches to predict energy consumption in buildings: engineering methods that utilize physically-based simulations; statistical methods that relate energy use with the associated design parameters via relevant data; artificial intelligence algorithms that perform predictions using the related learning data; and hybrid or grey-box methods that combine white- and black-box methods to eliminate their shortfalls. The review evaluated those methods against model complexity, running speed, ease of use, inputs complexity and accuracy. Similarly, Fouquier et al. (2013) highlighted three approaches to predict energy consumption: physically-based white-box that entails computational fluid dynamics, besides zonal and nodal methods; statistical black-box that includes

Multiple Linear Regression (MLR), Artificial Neural Network (ANN), Support Vector Machine (SVM) and Genetic Algorithms (GA); and grey-box methods. Each method was discussed in terms of applications, complexity, ease of use, computation time, data size and algorithm hyperparameters. Ahmad et al. (2014) outlined two approaches to predict energy use: MLAs, such as ANN and SVM; and hybrid methods. They were compared against complexity, ease of use, speed, inputs type and accuracy.

An interesting review by Fumo (2014) summarized different attempts to classify energy prediction, focusing on whole building energy, model verification and weather data. The review grouped four approaches to predict energy: hybrid; statistical; engineering; and steady-state or dynamic methods, acknowledging that the best approach would depend on type, scale and size of data, besides design stage. Wang and Srinivasan (2017) identified three categories of energy prediction: white-, black- and grey-box approaches, focusing on black-box single algorithms, such as MLR, ANN and Support Vector Regression (SVR); besides other ensemble algorithms (Hansen and Salamon, 1990). The review compared those approaches against ease of use, applications, accuracy, computation speed and difficulty, providing suggestions for future trends. Another review on the use of black-box approaches to predict energy consumption in buildings was conducted by Amasyali and El-Gohary (2018), who cited 63 academic research papers, and presented a comparison of the used MLAs against building type, temporal graduality, type of energy consumption, purpose of prediction, data type, data size, data pre-processing, feature type and performance evaluation. The review also discussed the related future research directions. Likewise, Wei et al. (2018) compared three approaches to predict energy: white-, grey- and black-box, along with their applications. The review focused on black-box method, and reviewed 61 research papers that utilized ANN, SVM, MLR, Decision Tree (DT), GA, K-means, Self-Organizing Map and Hierarchical Clustering, offering a comparison among them against scale, measure length, input variables and data source.

The accumulations of those review studies mainly identified three approaches to predict building performance: white-, black- and grey-box (Fig. 1), confirming the focus that was given to MLAs to solve regression, classification and clustering problems. Regarding the black-box approach, which is of main concern to this review, the used criteria and considerations to compare MLAs themselves can be realized by algorithm-related features that involve, according to previous reviews: scope of prediction, applications, available data, complexity, ease of use, speed, accuracy and hyperparameters. Herein, choosing the best algorithm mainly depends on case-specific, data-related attributes, which include the available data type, scale and size.

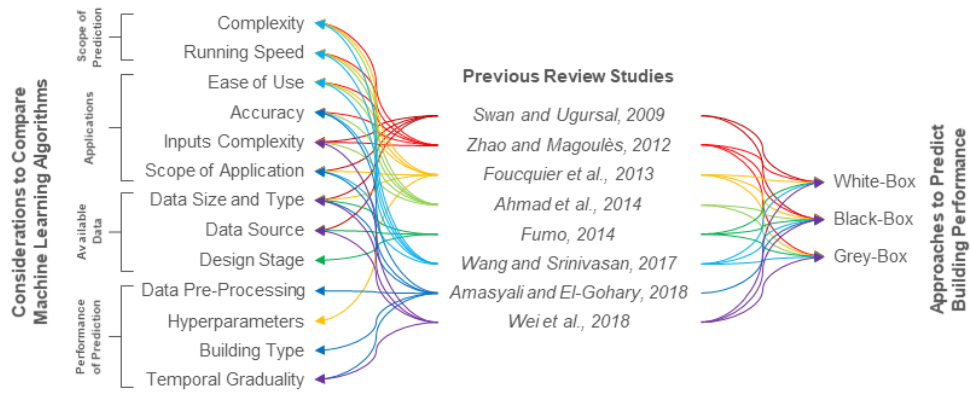


Fig. 1. The reviewed approaches to predict building performance with different criteria to compare MLAs.

3. Research Aim and Methodology

To the author's best knowledge, hardly any attempt can be found in literature that addresses reviewing previous research attempts to use MLAs to predict luminous conditions and daylighting performance in buildings. Such scarcity can be attributed to the relatively novel, yet promising, field of application. Thus, this research aims to present a review on the growing directions found in the scholarly literature that exploit different MLAs to rapidly predict daylighting performance inside buildings. This review will only consider MLAs that offer accelerated approximations, rather than other optimization algorithms that take a longtime to find global optima. Also, it will not reflect the research landscape of neither electrical lighting nor external daylighting, as they are out of the scope, and cannot fit the extent of its specified target; yet, a study that considers those issues can be undertaken separately in the future.

The work of this review builds on previous reviews on different machine learning approaches to predict energy consumption in buildings (Ahmad et al., 2014; Amasyali and El-Gohary, 2018; Fouquier et al., 2013; Fumo, 2014; Swan and Ugursal, 2009; Wang and Srinivasan, 2017; Wei et al., 2018; Zhao and Magoulès, 2012). It is based on a thorough analysis and citation of up-to-date academic research papers, considering a set of criteria to compare MLAs, derived from previous review studies (Fig. 1). They include four main groups, each of which comprises a number of subcategories: (i) Scope: building types, climate zones and locations; (ii) MLAs: problem type, selected MLAs with their hyperparameters, input parameters and output parameters; (iii) Data: data sources, data sizes and temporal granularities; as well as (iv) Accuracy: evaluation metrics. Finally, the review offers a detailed discussion about those criteria. This review seeks to be more inclusive, yet easier to be realized by architects of different backgrounds and expertise. That should increase the applicability to predict internal daylighting using machine learning, and consequently, support architects' design decision-making. Another implication of this work is to point towards knowledge gaps and missing opportunities in the related research domain, revealing future trends and perspectives that allow for such innovative approaches to be exploited more commonly in Architectural practice. It is expected that, sooner than later, machine learning will contribute to more challenging problems in Architecture, and specifically, in the field of daylighting prediction.

To achieve the research aim, the methodological procedure is realized through several steps. First, a Boolean keyword search is conducted on the scholarly literature by Google Scholar and Egyptian Knowledge Bank, allowing to combine the related keywords with search operators such as 'AND' and 'OR' to find detailed results. Such engines facilitate a broad search across many disciplines and sources within accredited academic databases and publishers. To focus on the related research domain and the targeted scope, the resulting scholarly literature is screened and analytically reviewed to identify their purpose and scope. Each of which is then filtered to distinguish relevant candidate articles within the scope of this review. In the refined scholarly literature, the articles that were cited by other articles are filtered too using the same relevance criteria. Those that passed the filtration are also identified as additional candidate articles. All candidate articles are then categorized as per the topical structure of the entire review, which is divided into the following sections:

- Section 1 provides a brief introduction on white- and black-box approaches;
- Section 2 highlights the related review studies that focused on the use of MLAs in building performance, defining their scope;
- Section 3 defines the research aim and describes the methodological procedure of this review;
- Section 4 describes machine learning, its process and categories, along with a brief background on some of the famous MLAs;
- Section 5 analyzes previous studies that utilized MLAs to predict internal daylighting conditions, in terms of the identified four criteria groups: Scope of Prediction; MLAs; Data Sources and Sizes; and Evaluation Metrics;
- Section 6 discusses those studies in detail considering the criteria and their subcategories to reveal the missing opportunities in the related research domain;
- Section 7 directs future research perspectives and trends; and
- Section 8 concludes the review findings.

4. Background

4.1. Machine Learning Process and Categories

Derived from computer science, machine learning is the study of statistical models and algorithms that use a dataset's variables to recognize relevant spatiotemporal patterns and information. Machine learning offers computer systems with whole new abilities, as they can perform rapid predictions from newly input data, without being instructed to perform such task (Mitchell, 1997). Basically, machine learning searches sample training data to build and formulate a mathematically-fit model (Kalogirou, 2001; Liu et al., 2019), which maps the complex relationships between independent inputs and target outputs. This is where evaluation metrics are used to assess the model's performance (Li and Lou, 2018). Machine learning process encompasses a set of steps: (i) Data Collection entails gathering relevant historical, experimental, observational, or simulation-derived data, before constructing the model. The quality of data directly affects how accurate the predictive model will be. (ii) Data Preparation involves data normalization, scaling and randomization; and this is essential, as the ranges of data variables can vary widely, influencing how MLAs function. Data randomization is conducted as well, to

prevent the order of data from affecting the learning. (iii) Data Exploration and Visualization by means of statistical and visualization techniques is vital to test if there are relevant relationships between input variables, and to detect any data imbalances so that the developed model will not be biased towards predicting a particular range of variables. (iv) Data Pre-Processing includes splitting the data into 3 parts. The first, and the largest, is used for training the model; the second is for testing or evaluating the performance of the model during the training; and the third part is for validating or fine-tuning MLAs hyperparameters' to offer further improvements to the model's performance. Finally, (v) Prediction are then conducted, where the worth of machine learning is realized accordingly.

MLAs differ in the used technique, the type of variables to handle, and the nature of tasks to solve. Still, they can be grouped according to the learning technique into 3 categories: supervised, unsupervised, and reinforcement (Bishop, 2006) (Fig. 2); each of which has its strengths and weaknesses. Supervised learning analyzes possibly generalized dataset to build the model that links input variables with output ones, based on training dataset of many input-output pairs. The model can then predict target variables from newly input data. For classification and regression, the supervised learning algorithms include MLR, ANN, SVM, Naïve Bayes (NB), DT and Random Forest (RF). Unsupervised learning detects unknown patterns in a dataset to recognize target variables without the output data. The model realizes unsorted information based on similarities, differences and patterns. For clustering, feature selection and dimensionality reduction, the unsupervised learning algorithms include Hierarchical Clustering, K-means and Principal Component Analysis (PCA). Reinforcement learning pursues a specific target in a dynamic environment without knowing if the model converges to the target or not. It takes an action as input, and generates a maximum expected reward as output, where the process is driven by the feedback from the environment. Reinforcement learning is studied in many disciplines, such as operations research, swarm intelligence and GA. Well-established models would require adequate and unbiased databases that include wide ranges of inputs and outputs for training, validating and testing (Basheer and Hajmeer, 2000; Kaytez et al., 2015).

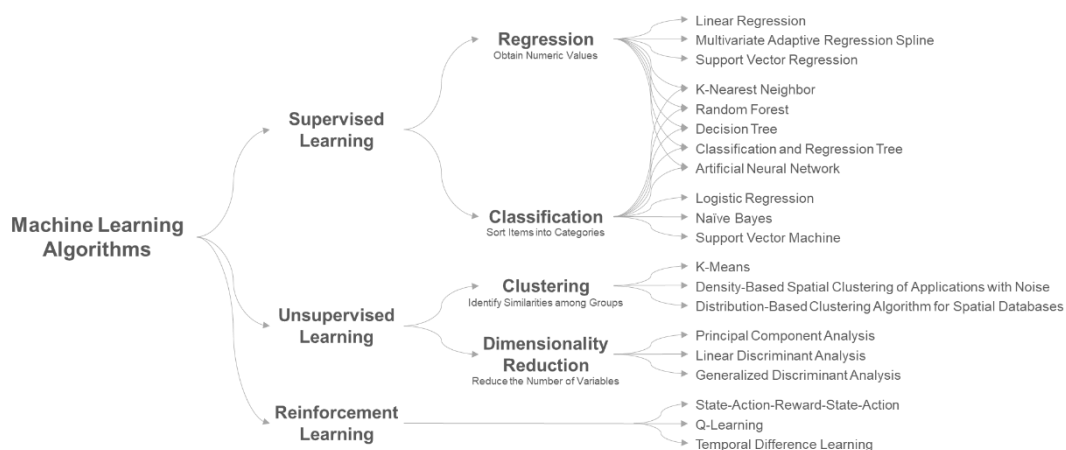


Fig. 2. Different categories of MLAs according to learning technique, with examples.

4.2. Artificial Neural Network

ANN is a non-linear function approximation algorithm that basically performs numerical predictions via supervised, unsupervised and reinforcement learning techniques (Yao, 1999). Other ANN variations are used for classification, clustering and pattern recognition (Jain et al., 1996). The biological structure of neural networks that constitutes the brain and nervous system is the main concept behind ANN. Essentially, ANN is composed of a number of layers; each of which contains several processing elements, called neurons. The input layer receives the input data; the hidden layer(s) performs most of the computations; while the output layer predicts the results (Hecht-Nielsen, 1992) (Fig. 3). Neurons of one layer are interconnected to other neurons in the consequent layer by connections that are assigned with weights (Jain et al., 1996). Inputs (x_i) are multiplied by corresponding weights (w_{ij}), and the weighted sum is adjusted by bias (θ_i) to minimize the error that stems from the difference between the actual and predicted outputs. Then, this value passes through an activation function (f_i), such as Unit Step, Sigmoid, Piecewise Linear and Gaussian (Bishop, 2006), which acts as a gate that determines if the neurons should transmit new data to the next layer or not. The training of ANN entails tuning the weights in the neurons depending on the outputs. According to the flow of information and the connection weights, data is propagated throughout the whole network; and when it reaches the output layer, the neuron with highest value defines the output. This is where the difference between the predicted and actual outputs is attained, changing back the connection weights to minimize that difference. Such Back-Propagation training process (Rumelhart and Hintonf, 1986) is iteratively updated until a termination condition is met. Readers can refer to (Schmidhuber, 2015) for a historical overview and additional features on ANN.

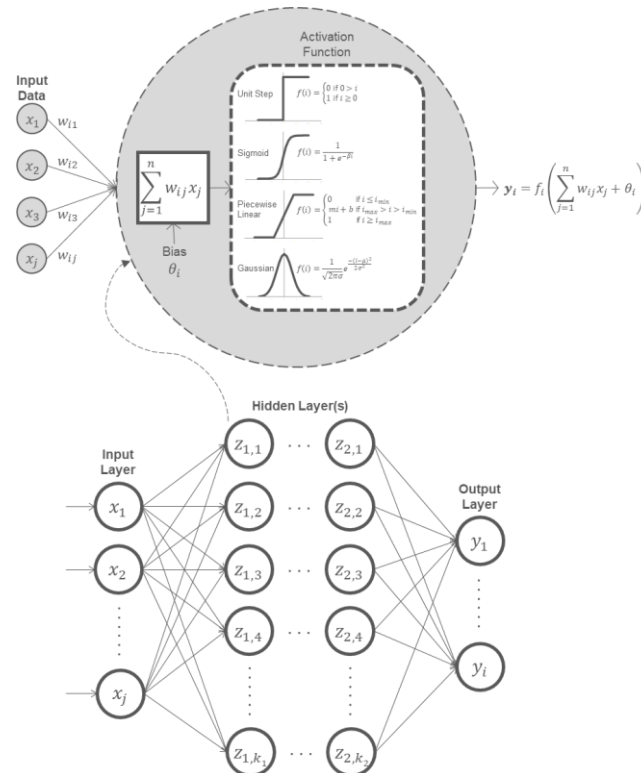


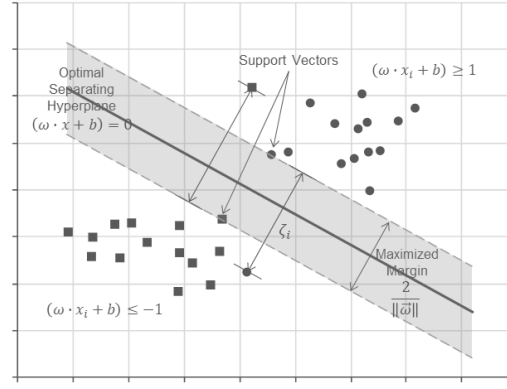
Fig. 3. An example of ANN architecture, showing different layers, their corresponding weights, and various types of activation functions that can be applied to the weighted sum.

Key hyperparameters to tune ANN before training include: the number of hidden layers; the number of hidden neurons; the learning rate that defines the update frequency to adjust the errors; the number of epochs that identifies the number of iterations such ANN undergoes for the whole training dataset; the batch size to control the number of propagated training samples through the network before updating the model; and the activation function that introduces non-linear properties to ANN and converts inputs of neurons to outputs. Variations of ANN include Feed-Forward Neural Network (FFNN) (Ivakhnenko, 1971), Back-Propagation Neural Network (BPNN) (Rumelhart and Hintonf, 1986), Radial Basis Function Neural Networks (RBFNN) (Broomhead and Lowe, 1988), Recurrent Neural Networks (RNN) (Elman, 1990), Adaptive Neuro-Fuzzy Inference System (ANFIS) (Jang, 1993) and Convolutional Neural Networks (CNN) (LeCun et al., 1998).

4.3. Support Vector Machine

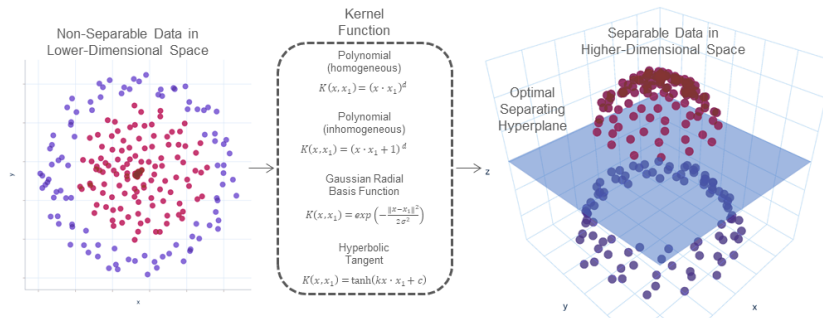
SVM is a discriminative classifier that is used fundamentally for classification, pattern recognition and regression by supervised learning techniques (Boser et al., 1992). The concept behind SVM is to differentiate between groups of linearly separable data features, called vectors, which belong to different categories (Fig. 4). This is where an optimal separating hyperplane is defined geometrically by the nearest vectors, called support vectors. The hyperplane should be as far as possible from those support vectors; thus, SVM seeks to maximize the decision boundary, or the margins (ω), of such hyperplane. This will ensure the accuracy of classification, allowing for more space for newly input vectors that can be classified according to the hyperplane side they belong (Wu et al., 2008). To this point, the data is assumed to be linearly separable, where it is possible to find a separating hyperplane that will perfectly classify the data; but this rarely occurs in real-life. Therefore, instead of letting SVM classify data correctly and risk overfitting, some mistakes are intentionally allowed by introducing a penalty for each data vector, the slack variable (ξ). In this sense, SMV seeks to lessen the slack variables by minimizing the sum of their distances from the margins, while maximizing the hyperplane's margins themselves.

Yet again, this may be too simple for practical applications; thus, SVM can handle complex non-linear problems by transforming the data features into a higher-dimensional space using a non-linear kernel function, such as Polynomial, Gaussian Radial Basis Function (RBF) and Hyperbolic Tangent (Aiserman et al., 1964). In a (d) dimensional features space, the hyperplane is a (d-1) dimensional separator; for instance, in a 3D space, the hyperplane is a 2D plane. As the separating hyperplane of maximized margin is deployed, different vectors can be classified as linearly separable data in such higher-dimensional space (Fig. 4) (Vapnik, 1999). Then, the non-linear solution is implicitly attained by projecting the results to the original lower-dimensional space (Kotsiantis, 2007). For further details and overview on SVM, readers may refer to (Burges, 1998). Essential hyperparameters to tune SVM include: parameter (C) that defines the margin width for penalizing the misclassified vectors to avoid overfitting; the used kernel function; and the kernel's specific parameters, such as the degree of Polynomial Kernel and the width of RBF Kernel. Other diversifications of SVM include Least-Squares Support Vector Machine (LS-SVM) (Suykens and Vandewalle, 1999), Support Vector Clustering (SVC) (Ben-Hur et al., 2001), Bayesian SVM (Polson and Scott, 2011) and SVR (Drucker et al., 1997).



For Linear SVM, the optimization problem will be:

$$\text{Min } (\omega, b) \left\{ \frac{\|\omega\|^2}{2} + C \sum_{i=1}^N \zeta_i \text{ subject to } y_i(\omega \cdot x_i + b) \geq 1 - \zeta_i \text{ for any } i = 1, \dots, N \right.$$



For Non-Linear SVM, the optimization problem will be:

$$\text{Min } (\omega, b) \left\{ \frac{\|\omega\|^2}{2} + C \sum_{i=1}^N \zeta_i \text{ subject to } y_i(\omega \cdot \phi(x_i) + b) \geq 1 - \zeta_i \text{ for all } 1 \leq i \leq N, \zeta_i \geq 0 \right.$$

Fig. 4. An example of linear and non-linear SVM, showing the definition of optimization problems and the optimal separating hyperplane in both cases.

4.4. Other Popular Algorithms

Other algorithms, including Regression, DT and RF, have received much recognition, and besides daylighting, they were implemented to predict different aspects of building performance. Regression entails a set of methods that mainly predict the relationship between two sets of variables: a response (dependent) variable and explanatory (independent) variables. As an inherently supervised algorithm, MLR is one of the most exploited form of regression, by which a linear equation fits the training dataset, making assumptions about the linearity, variance and independence of errors. There are many examples of Regression algorithms designed for different purposes according to the data type and distribution, such as Polynomial Regression (Smith, 1918), Logistic Regression (Cramer, 2002; Kleinbaum et al., 2002), Bayesian Regression (Fienberg, 2006) and Auto Regression (AR) (Akaike, 1969).

DT is a supervised algorithm that is widely used to solve both classification and regression problems. For a given set of inputs with specific features, DT maps an inverted treelike structure that consists of nodes and branches to predict the output that results from hierarchical steps of decisions. DT starts with a root node that checks if the prediction of a given input values or classes is satisfied. This is where the data is split into branches that connect the nodes, representing yes or no decision responses. Then, each internal (non-leaf) node evaluates the next input feature; and the decisions continue to traverse

downwards, leading eventually to a leaf node that corresponds to the final prediction output. Such DT structure can grow flexibly according to the input data. There are many diversifications of DT algorithm, which include Iterative Dichotomiser 3 (ID3) (Quinlan, 1986), C4.5 (Quinlan, 1993), Classification and Regression Tree (CART) (Breiman et al., 1984), RF (Ho, 1995) and Chi-Square Automatic Interaction Detection (CHAID) (Kass, 1980).

All those MLAs rely on supervised learning approach, and can be used for classification and regression problems considering discrete or continuous data. They received much attention by many researchers; and the next section reviews previous studies that utilized MLAs to predict luminous conditions and daylighting performance inside buildings.

5. Previous Studies on Machine Learning Algorithms to Predict Internal Daylighting

The landscape of scholarly literature that put the focus on employing different MLAs to predict internal daylighting conditions is limited, yet continues to grow. To help architects comprehend the most recent directions, while offering guidance for future research trends and areas of development as well, a total of 27 related studies, which were published in the last 13 years (2006-2019), were found, then reviewed thoroughly. Every study involves a number of investigational attempts that vary according to the objective, building type, location, selected algorithm and output parameters. Consequently, the total number of investigational attempts exceeds the number of studies themselves (109 attempts for 27 research paper). (Table 1) presents a comprehensive outline on such recent work, which is arranged chronologically in terms of: Scope of Prediction, MLAs, Data Sources and Sizes, and Evaluation Metrics; each of which is discussed in the following section with the same order. The summary of the coordinates plot's results and the distributions of their frequency rates are shown in (Fig. 5) according to: (a) Building Type, (b) Climate Zone, (c) Location, (d) Problem Type, (e) Selected MLAs, (f) Hyperparameters, (g) Input Parameters, (h) Output Parameters, (i) Data Source, (j) Simulation Tool, (k) Temporal Granularity and (l) Error Metrics.

Table 1. Summary of the reviewed researches that considered predicting internal daylighting using machine learning, in terms of scope, the used algorithm, data and accuracy.

Author(s)	Scope		Machine Learning Algorithm					Data			Accuracy
	Building Type	Location and Climate Zone	Problem Type	Selected Algorithm	Hyperparameters	Input Parameters	Output Parameters	Data Source	Data Size	Temporal Granularity	Evaluation Metrics
(Kurian et al., 2006)	N/S	Manipal, Karnataka, India (Tropical Monsoon, Am)	Regression	ANN	No. Epochs	Window Width Window Height Windows Location Room Width Room Length Room Height Room Elevation	Illuminance Values	Simulation-Derived (Radiance)	500 for Training 500 for Testing	Annual Timeseries (Hourly Basis)	0.01-3.50 lx (PE) 0.13-1.12% (RMSE)
					AR	N/S					0.19-0.91 lx (PE)
(Binol, 2008)	Office	Izmir, Turkey (Hot-Summer Mediterranean, Csa)	Regression	ANN	No. Hidden Neurons No. Epochs	Orientation Sensor Point Identification Distance from Windows Floor/Room Identification Number of Windows Room Width Room Length Room Height Global Horizontal Radiation Relative Humidity UV Index UV Dose Time of the Day Day of the Year	Illuminance Values	Field Measurements	3,168 for Training 792 for Testing	Selected Instances (Hourly Basis)	2.20-2.54% (PE)
(Conraud-Bianchi, 2008)	Educational	Grono, Norway (Subarctic, Dfc)	Regression	ANN	No. Hidden Neurons	Orientation Window Width Window Height Shading Device Shading Parameters Dry Bulb Temperature	DF	Simulation-Derived (ESP-r)	1,200 for Training 300 for Testing	Annual (Hourly Basis)	0.72-2.50% (ARE)
(Kurian et al., 2008)	N/S	Mangalore, Karnataka, India (Tropical Monsoon, Am)	Regression	ANN	No. Epochs	Orientation Direct Normal Radiation Diffuse Horizontal Radiation Dry Bulb Temperature Time of the Day	Illuminance Values	Simulation-Derived (Ecotect)	500 for Training 500 for Testing	Annual Timeseries (Hourly Basis)	0.01-3.50 lx (PE) 0.13-1.12% (RMSE)
					AR	N/S					N/S
(Kazanas et al., 2009)	Office	Izmir, Turkey (Hot-Summer Mediterranean, Csa)	Regression	ANN	No. Hidden Neurons	Orientation Sensor Point Identification Distance from Windows Floor/Room Identification Number of Windows Room Width Room Length Room Height Global Horizontal Radiation Dry Bulb Temperature Relative Humidity UV Index UV Dose Time of the Day Day of the Year	Illuminance Values	Field Measurements	8,000 for Training 2,000 for Testing	Selected Instances (Hourly Basis)	1.08-3.32% (PE)
(Ahmed et al., 2011a)	Office	Cork, Ireland (Temperate Oceanic, Cfb)	Classification	NB	N/S	Orientation Window Width Window Height Room Width Room Length Room Height Room Elevation	Illuminance Values	Field Measurements Simulation-Derived (Ecotect)	906,540 for Training 140,218 for Testing	Selected Instances (Sub-Hourly Basis)	99.47% (PC)
					DT	N/S					99.39% (PC)
	Laboratory					Wall Material Floor Material Ceiling Material Obstruction Height					

				SVM	N/S	Obstruction Length Obstruction Width Global Horizontal Radiation Global Horizontal Illuminance					99.72% (PC)
(Ahmed et al., 2011b)	Office Laboratory	Cork, Ireland (Temperate Oceanic, Cfb)	Regression	DT	N/S	Orientation Window Width Window Height Room Width Room Length Room Height Room Elevation Work Plane Height Wall Material	Illuminance Values	Field Measurements Simulation-Derived (Ecotect)	906,540 for Training 140,218 for Testing	Selected Instances (Sub-Hourly Basis)	99.39% (PC)
				SVM	N/S	Floor Material Ceiling Material Obstruction Height Obstruction Length Obstruction Width Global Horizontal Radiation Diffuse Horizontal Radiation Time of the Day					94.99% (PC)
(Hu and Olbina, 2011)	Office	Gainesville, Florida, US (Humid Subtropical, Cfa)	Regression	ANN	No. Hidden Layers No. Hidden Neurons	Global Horizontal Radiation Direct Normal Radiation Diffuse Horizontal Radiation Horiz. Infrared Radiation Intensity Global Horizontal Illuminance Diffuse Horizontal Illuminance Exterior Horiz. Beam Illuminance Dry Bulb Temperature Relative Humidity Time of the Day Solar Altitude Solar Azimuth Solar Declination Solar Hour	Illuminance Values	Simulation-Derived (DElight)	1,404 for Training 936 for Testing	Selected Instances (Hourly Basis)	1.50-5.30% (PE)
(da Fonseca et al., 2013)	Office	Florianópolis, Brazil (Humid Subtropical, Cfa)	Regression	ANN	No. Hidden Neurons No. Epochs Learning Rate Activation Function	Orientation Window Width Window Height Glazing Properties Shading Device Shading Parameters	Illuminance Values	Simulation-Derived (DAYSIM)	195 for Training 21 for Testing	N/S	0.03% (MSE)
				MLR	N/S						0.05% (MSE) 0.80 (R ²)
(Inanici, 2013)	N/S	Seattle, Washington, US (Warm-Summer Mediterranean, Csb)	Regression	LSMR	N/S	Orientation Global Horizontal Illuminance Internal Illuminance Pixel Luminance Time of the Day Day of the Year Solar Altitude Solar Azimuth	Illuminance Values	Field Measurements (High Dynamic Range Imagery)	50 for Training 50 for Testing	Selected Instances (Sub-Hourly Basis)	0.69-0.99 (R ²)
								Simulation-Derived (Radiance)			0.87-0.90 (R ²)
(Colaco et al., 2014)	N/S	Manipal, Karnataka, India (Tropical Monsoon, Am)	Regression	ANN	No. Hidden Layers No. Hidden Neurons	Sensor Point Identification Glazing Properties Wall Material	Illuminance Values	Field Measurements	49,180 for Training 21,080 for Testing	Selected Timeseries (Sub-Hourly Basis)	2.20-4.60% (RMSE) 0.98 (R ²)
				AR	N/S	Floor Material Ceiling Material					7.50-9.20% (RMSE) 0.94-0.96 (R ²)
(Logar et al., 2014)	Office	Ljubljana, Slovenia (Temperate Oceanic, Cfb)	Regression	ANN	No. Membership Functions	Shading Device Shading Parameters Global Horizontal Radiation Diffuse Horizontal Radiation Global Horizontal Illuminance	Illuminance Values	Field Measurements	42,000 for Training 28,000 for Testing	Selected Timeseries (Sub-Hourly Basis)	12.60-18.10% (RMS) 7.76-12.2% (MBE)
(Liu et al., 2015)	Industrial	City of Singapore, Singapore (Tropical Rainforest, Af)	Clustering	DBSCAN	N/S	Occupancy Schedule Glazing Properties Room Width Room Length Room Height Wall Material Floor Material Ceiling Material Shading Device Shading Parameters	DA	Simulation-Derived (Radiance)	N/S	Annual (Hourly Basis)	N/S

(Zhou and Liu, 2015)	Office	Virginia, US (Humid Subtropical, Cfa)	Classification	ANN	No. Hidden Layers	Occupancy Schedule Sensor Point Identification Window Width Window Height Windows Location Glazing Properties Wall Material	UDI	Simulation-Derived (OpenStudio)	729 for Training 729 for Testing	Annual (Hourly Basis)	with PCA: 3.65-7.49% (PE) without PCA: 9.71-16.35% (PE)
				SVM	Kernel Functions	Floor Material Ceiling Material Diffuse Horizontal Illuminance Exterior Horiz. Beam Illuminance Solar Altitude Solar Azimuth					with PCA: 29.86-37.85% (PE) without PCA: 26.12-29.92% (PE)
(Chatzikonstantinou and Sariyildiz, 2016)	Office	Izmir, Turkey (Hot-Summer Mediterranean, Csa)	Regression	ANN	Activation Function No. Hidden Layers No. Hidden Neurons	Orientation Occupancy Schedule Sensor Point Identification	DA DGP	Simulation-Derived (Honeybee)	2,000 for Training 500 for Testing	Annual (Hourly Basis)	0.07-0.32% (RMSE) 0.68-0.81 (R ²)
				SVM	Kernel Functions (C) Parameter Width of RBF Kernel	View Direction Window Width Window Height Windows Location					0.08-0.13% (RMSE) 0.59-0.94 (R ²)
				RF	No. Trees No. Features	Room Width Room Length Room Height					0.07-0.16% (RMSE) 0.66-0.91 (R ²)
				k-NN	N/S	Room Height Time of the Day Day of the Year					0.09-0.14% (RMSE) 0.38-0.92 (R ²)
(Navada et al., 2016)	Office	Manipal, Karnataka, India (Tropical Monsoon, Am)	Regression	ANN	No. Hidden Layers No. Hidden Neurons No. Epochs	Room Width Room Length Room Height Work Plane Height Shading Device Shading Parameters Global Horizontal Radiation Direct Normal Radiation Diffuse Horizontal Radiation	Illuminance Values	Field Measurements	5,696 for Training 2,848 for Testing	Selected Timeseries (Sub-Hourly Basis)	0.97-5.24% (PE)
(Ahmad et al., 2017)	Educational	Cardiff, Wales, UK (Temperate Oceanic, Cfb)	Regression	ANN	No. Hidden Layers No. Hidden Neurons	Occupancy Schedule Shading Device Shading Parameters Global Horizontal Radiation Direct Normal Radiation Diffuse Horizontal Radiation	Illuminance Values	Simulation-Derived (EnergyPlus)	N/S	Annual (Hourly Basis)	278.49-282.35 (RMSE) 0.98 (R ²)
				RF	No. Trees No. Features	Dry Bulb Temperature Relative Humidity Wind Speed Time of the Day Day of the Year Solar Altitude Solar Azimuth					227.87-235.738 (RMSE) 0.97-0.98 (R ²) 44.91-86.80 (MAD)
(Lorenz and Jabi, 2017)	Office	N/S	Regression	ANN	No. Hidden Layers No. Hidden Neurons	Orientation Sensor Point Identification Distance from Windows Floor/Room Identification Room Width Room Length Room Height Work Plane Height Wall Material Floor Material Ceiling Material	DA	Simulation-Derived (DIVA-for-Rhino)	270 for Training 30 for Testing	Annual (Hourly Basis)	0.11-0.28% (MSE)
(Nault et al., 2017)	Office	Geneva, Switzerland (Temperate Oceanic, Cfb)	Regression	MLR	N/S	Orientation Window Width Window Height Windows Location Glazing Properties Room Width Room Length Room Height Room Elevation	SDA	Simulation-Derived (DIVA-for-Rhino)	280 for Training 280 for Testing	Annual (Hourly Basis)	2.82-7.14% (RMSE) 0.59-0.98 (R) 4.30-20.90% (PE)

						GP	Obstruction Height Obstruction Length Obstruction Width Obstruction Angle Distance from Obstruction Global Horizontal Radiation Direct Normal Radiation Diffuse Horizontal Radiation					3.16-21.09% (RMSE) 0.48-0.97 (R) 33.40-48.90% (PE)
(Uribe et al., 2017)	Office	Antofagasta, Chile (Hot-Desert, BWh) Santiago, Chile (Warm-Summer Mediterranean, Csb) Punta Arenas, Chile (Subpolar Oceanic, Cfc)	Regression	ANN	N/S		Window Width Window Height Glazing Properties Wall Material Floor Material Ceiling Material Shading Device Shading Parameters	sDA ASE	Simulation-Derived (Groundhog)	5400 for Training	Annual (Hourly Basis)	N/S
(Verso et al., 2017)	Office	Berlin, Germany (Temperate Oceanic, Cfb) Turin, Piemonte, Italy (Humid Subtropical, Cfa) Catania, Sicily, Italy (Hot-Summer Mediterranean, Csa)	Regression	MLR	N/S		Orientation Window Width Window Height Windows Location Glazing Properties Room Length Work Plane Height Obstruction Height Obstruction Angle Shading Device Shading Parameters	DA CDA sDA	Simulation-Derived (DAYSIM)	510 for Training 48 for Testing	Annual (Sub-Hourly Basis)	2.76-13.92% (RMSE) 0.03-0.11 (CV)
(Yacine et al., 2017)	Educational	Biskra, Algeria (Hot-Desert, BWh)	Regression	ANN	No. Hidden Neurons		Internal Illuminance	DGP CGI	Field Measurements (Experimental Survey)	59 for Training 31 for Testing	Selected Instances (Hourly Basis)	39.20-69.60% (RER)
(Beccali et al., 2018)	Laboratory	Palermo, Italy (Hot-Summer Mediterranean, Csa)	Regression	ANN	N/S		Sensor Point Identification Global Horizontal Radiation Global Horizontal Illuminance Time of the Day Day of the Year Solar Altitude Solar Azimuth	Illuminance Values	Field Measurements	64,786 for Training 11,432 for Testing	Selected Instances (Sub-Hourly Basis)	0.22-0.47% (MSE) 0.91-0.95 (R ²)
(Liu et al., 2018)	N/S	Seattle, Washington, US (Warm-Summer Mediterranean, Csb)	Regression	ANN	N/S		Pixel Location Room Width Room Length Room Height Wall Material Floor Material Ceiling Material Direct Normal Radiation Diffuse Horizontal Radiation Global Horizontal Illuminance Pixel Luminance Solar Altitude Solar Azimuth	Illuminance Values	Simulation-Derived (Radiance)	3,879 for Training 500 for Testing	Annual (Hourly Basis)	0.01072-0.06389% (MSE) 0.01 (RER)
(Lorenz et al., 2018)	N/S	N/S	Regression	ANN	No. Hidden Neurons		Sensor Point Identification Distance from Windows Window Width Window Height Windows Location Obstruction Height Distance from Obstruction	DA	Simulation-Derived (DIVA-for-Rhino)	2057 for Training 685 for Testing	Annual (Hourly Basis)	0.51-0.55% (MAE) 0.63-0.65% (RMSE) 0.06-0.09% (MBE)
(Radziszewski and Waczińska, 2018)	Office	Denver, Colorado, US (Cold Semi-Arid, BSk)	Regression	ANN	No. Hidden Layers No. Hidden Neurons No. Epochs Learning Rate Activation Function		Orientation Window Width Window Height Windows Location Room Width Room Length Room Height Time of the Day Day of the Year	DF DA DGP	Simulation-Derived (DIVA-for-Rhino)	2963 for Training 200 for Testing	Annual (Hourly Basis)	0.20-0.80% (MSE)

(Ayoub, 2019a)	Residential	Alexandria, Egypt (Hot-Desert, BWb)	Regression	MLR	N/S	Window Width Window Height Windows Location Glazing Properties Obstruction Height Obstruction Angle Obstruction Material Distance from Obstruction	sDA ASE	Simulation- Derived (DIVA- for-Rhino)	1296 for Training	Annual (Hourly Basis)	5.28-8.13% (RMSE) 0.87-0.98 (R ²)
N/A: Not Available N/S: Not Specified											

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5.1. Scope of Prediction

5.1.1. Building Types

The Scope of Prediction can be categorized into building types, climate zones and locations. Regarding the building types, 5 categories are identified in the reviewed studies (Office, Educational, Laboratorial, Industrial and Residential). (Fig. 5-a) summarizes their distribution, indicating that 60% of the studies focused on office buildings, 11% and 6% considered educational and laboratorial spaces respectively, 2% included industrial and residential spaces equally, but the remaining 19% did not specify the studied building type. Of the reviewed studies that considered office spaces, 32% were performed in Temperate Oceanic climate, whilst 25% and 21% were conducted in both Hot-Summer Mediterranean and Humid Subtropical climates, respectively. On the problem type, 82% were used to solve regression problems, only 18% considered classifications, while clustering was not studied. About the used MLAs, 46% of the studies that considered studying office spaces relied on ANN, 18% used MLR, while 11% utilized both DT and SVM. Most of those studies that considered office spaces, 78%, depended on simulation tools to acquire training data, whereas the remaining 22% obtained data from field measurements. 57% of them covered long-term daylighting predictions, 36% selected certain temporal instances, but the remaining did not specify the used temporal granularity.

5.1.2. Climate Zones and Locations

In compliance with Köppen-Geiger classification (Critchfield, 1983; Geiger, 1954; Kottek et al., 2006), 10 climate zones are found in the reviewed studies (Temperate Oceanic, Hot-Summer Mediterranean, Tropical Monsoon, Humid Subtropical, Hot-Desert, Warm-Summer Mediterranean, Subarctic, Tropical Rainforest, Subpolar Oceanic and Cold Semi-Arid climates), as indicated in (Table 1). Those climate zones cover only 10 out of 30 sub-types of the updated Köppen-Geiger classification (Beck et al., 2018). This can be attributed to the small number of researches, where a researcher, or a group of researchers, conducted more than one investigation considering the same location, including (Ahmed et al., 2011a) and (Ahmed et al., 2011b) in Cork, Ireland; (Binol, 2008), (Kazanasmaz et al., 2009) and (Chatzikonstantinou and Sariyildiz, 2016) in Izmir, Turkey; (Kurian et al., 2006), (Colaco et al., 2014) and (Navada et al., 2016) in Manipal, Karnataka, India; besides (Inanici, 2013) and (Liu et al., 2018) in Seattle, Washington, US. Still, 69% of the reviewed studies focused on climate class C (temperate), while the remaining 18%, 11% and 2% included classes A (tropical), B (dry) and D (continental), respectively, but class E (polar) was not considered. (Fig. 5-b) reveals that Temperate Oceanic, Hot-Summer Mediterranean, Tropical Monsoon and Humid Subtropical climates are the most frequently used to predict internal daylighting, accounting for 28%, 17%, 15% and 13% of the attempts in the reviewed studies, respectively.

Those climate zones are representative of 23 locations around the world, as demonstrated in (Fig. 5-c) and (Fig. 6) (Cork, Ireland; Izmir, Turkey; Manipal, Karnataka, India; Seattle, Washington, US; Grong, Norway; Mangalore, Karnataka, India; Florianopolis, Brazil; Virginia, US; Cardiff, Wales, UK; Geneva, Switzerland; Biskra, Algeria; Gainesville, Florida, US; Ljubljana, Slovenia; City of Singapore, Singapore;

Antofagasta, Chile; Santiago, Chile; Punta Arenas, Chile; Berlin, Germany; Turin, Piemonte, Italy; Catania, Sicily, Italy; Palermo, Italy; Denver, Colorado, US; and Alexandria, Egypt).



Fig. 6. World map showing the distribution and the magnitudes of geographical locations that were considered in the reviewed studies.

5.2. Machine Learning Algorithms

5.2.1. Problem Types

The criterion of Machine Learning Algorithms can be realized through problem types, selected MLAs along with their hyperparameters, input parameters and output parameters. Regarding the problem types, 3 categories are recognized in previous studies (Regression, Classification and Clustering). (Fig. 5-d) summarizes their distribution, showing that 87% of the reviewed studies sought to solve regression problems, but only 11% and 2% of them examined classification and clustering, respectively.

5.2.2. Selected Machine Learning Algorithms and Hyperparameters

According to publishing year, (Fig. 7) reveals the distribution of attempts to use MLAs to predict internal daylighting in previous studies. Until 2012, only 4 MLAs have been used (ANN, SVM, AR, DT) to solve regression problems, in addition to one attempt to classify illumination categories using NB (Ahmed et al., 2011a). Later, 6 more MLAs were additionally employed to solve regression, classification and clustering problems (MLR, RF, LSMR, DBSCAN, k-NN and Gaussian Processes (GPs)). (Fig. 5-e) illustrates their distribution of application, where almost half of previous studies used ANN to predict internal daylighting, whereas 13%, 11%, 7% and 6% utilized MLR, SVM, AR and DT, respectively. As stated earlier, the research scope focuses on MLAs that offer accelerated approximations, rather than other optimization algorithms that take a longtime to find global optima; thus, only one MLA was cited from (Conraud-Bianchi, 2008), although ANN and GA have been actually used. Overall, 37% of the studies sought to compare two or more MLAs against prediction accuracy, speed, among other criteria. They include (Kurian et al., 2008, 2006) who compared ANN and AR; (Ahmed et al., 2011a, 2011b) who compared NB, DT and SVM; (da Fonseca et al., 2013) who compared ANN and MLR; (Colaco et al., 2014) who compared ANN and AR; (Zhou and Liu, 2015) who compared ANN and SVM; (Chatzikonstantinou and Sariyildiz, 2016) who compared ANN, SVM, RF and k-NN; (Ahmad et al., 2017) who compared ANN and RF; and (Nault et al., 2017) who compared MLR and GPs.

Considering the used hyperparameters, 35% of the reviewed studies either did not mention the method to tune the used MLAs before training begins, or did not use any hyperparameters at all. The remaining 65% considered 3 groups of hyperparameters according to the used MLA (ANN: number of Hidden Neurons, Hidden Layers, Epochs, Activation Function and Learning Rate; SVM: Kernel Functions, C Parameter and Width of RBF Kernel; RF: number of Trees and Features). (Fig. 5-f) demonstrates their overall distribution, and since ANN is the most used MLA, its hyperparameters are also the most frequently utilized, where number of Hidden Neurons, Hidden Layers, Epochs, Activation Function and Learning Rate were employed with frequency rates of 21%, 11%, 10%, 5% and 3%, respectively.

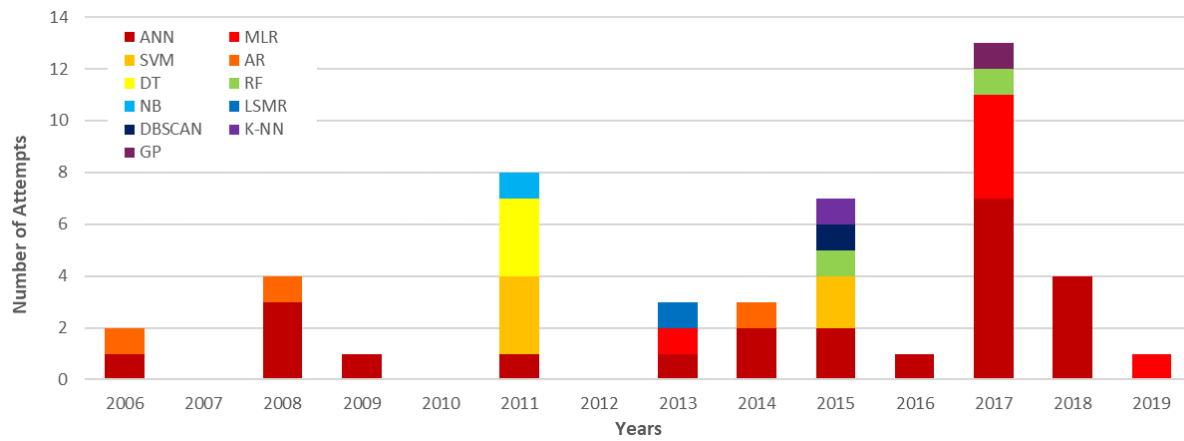


Fig. 7. Distribution of the previous attempts to use MLAs to predict internal daylighting according to publishing year.

5.2.3. Input Parameters

As mentioned before, MLAs can predict internal luminous conditions based on a set of input parameters. The reviewed studies show that the identified parameters are categorized into 2 major groups according to the input data: External and Internal; each of which can be subcategorized into 3 sets of parameters: (Climate Conditions; Temporal Settings; External Obstructions) and (Building Physical Features; Openings and Shading Devices; Occupancy and Sensor Data), respectively (Fig. 8). While the investigational purposes differ from study to another, a total of 48 input parameters can be identified that were selected to construct prediction models, based on the available data. (Fig. 5-g) shows their overall distribution.

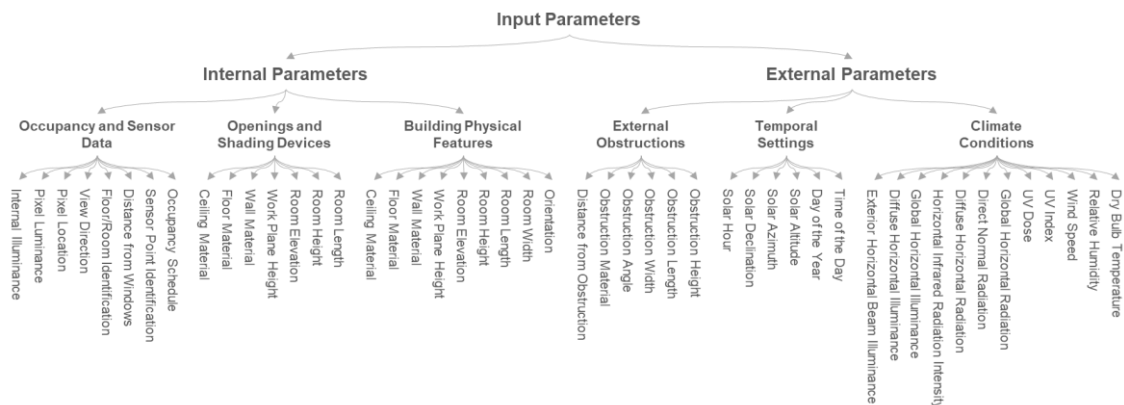


Fig. 8. The identified different sets of input parameters to predict internal daylighting

Regarding the (i) External Input parameters, (i-i) Climate Conditions include dry bulb temperature, relative humidity, wind speed, UV index, UV dose, global horizontal radiation, direct normal radiation, diffuse horizontal radiation, horizontal infrared radiation intensity from sky, global horizontal illuminance, diffuse horizontal illuminance and exterior horizontal beam illuminance. (i-ii) Temporal Settings include time of the day, day of the year, solar altitude, solar azimuth, solar declination and solar hour. (i-iii) External Obstructions include obstruction height, obstruction length, obstruction width, obstruction angle, obstruction material and distance from obstruction. Considering the (ii) Internal Input parameters, (ii-i) Building Physical Features include orientation, room width, room length, room height, room elevation, work plane height, wall material, floor material and ceiling material. (ii-ii) Openings and Shading Devices include window width, window height, windows location, number of windows, glazing properties, shading device and shading parameters. (ii-iii) Occupancy and Sensor Data includes occupancy schedule, sensor point identification, distance from windows, floor/room identification, view direction, pixel location, pixel luminance and internal illuminance.

In general, it can be concluded that the usage frequency of the external input parameters to perform predictions by the investigated attempts is 39%, while the frequency of the internal parameters is 61%. Specifically, Climate Conditions have a noteworthy impact on the internal daylighting. As they can easily be acquired from representative weather data files, they accounted for 19% of the used input parameters. Temporal Settings and External Obstructions, though important, represented only 11% and 8%, respectively. On the other hand, Building Physical Features, as they immediately affect the internal daylighting, comprised more than 30% of the input parameters, but Openings and Shading Devices, as well as Occupancy and Sensor Data, constituted 22% and 9%, respectively. Of all the input parameters, the most frequently used in the previous studies are: orientation, window width, window height, room width, room length, room height, global horizontal radiation, time of the day, diffuse horizontal radiation, windows location, sensor point identification, shading device, shading parameters, wall material, floor material, ceiling material and glazing properties.

5.2.4. Output Parameters

Considering the output parameters, they can be categorized into 2 main types: (Absolute Illuminance Values and Performance Metrics); both were predicted almost equally in previous studies, with frequencies of 53% and 47%, respectively. For clarity, the term 'Performance Metrics' is used in this review to signify daylight metrics, while 'Evaluation Metrics', which will be discussed later, denotes the accuracy measures used to assess how well MLAs models would perform. (Fig. 5-h) summarizes the overall distribution of output parameters, confirming that nearly 13%, 11%, 7%, 4%, 4%, 4%, 2% and 2% of the reviewed attempts focused on predicting DA, $sDA_{300/50\%}$, Daylight Glare Probability (DGP), Daylight Factor (DF), $ASE_{1000/250h}$, UDI, Continuous Daylight Autonomy (CDA) and CIE Glare Index (CGI), respectively. According to publishing year, (Fig. 9) reveals the distribution of previous attempts to predict daylighting in terms of absolute illuminance values and performance metrics.

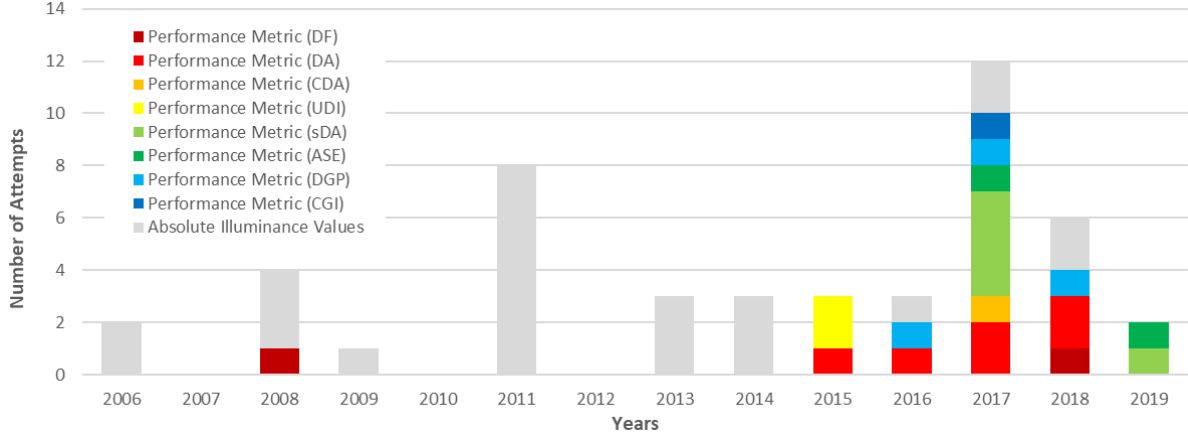


Fig. 9. Distribution of the previous attempts to predict daylighting in terms of absolute illuminance values and performance metrics according to publishing year.

With regard to daylight metrics, Trotter (1911) introduced DF to assess daylighting, which has been widely used since then. DF is the ratio between the received indoor illuminance (E_i) on a given plane to the corresponding level of unobstructed outdoor illuminance (E_o), excluding the direct sunlight from both illuminances (IES, 1972; Love, 1992), as follows:

$$DF = (E_i / E_o) \quad (1)$$

Later, DA was proposed by Association Suisse des Electriciens in 1989, which was later improved by Reinhart and Walkenhorst (2001). DA is the percentage of the annual occupied timesteps when the illuminance exceeds a predefined threshold:

$$DA = \frac{\sum_i (w f_i t_i)}{\sum_i t_i} \in [0,1], w f_i = \begin{cases} 1 & \text{if } E_{\text{Daylight}} \geq E_{\text{Limit}} \\ 0 & \text{if } E_{\text{Daylight}} < E_{\text{Limit}} \end{cases} \quad (2)$$

where (t_i) is the occupied hour of the year, ($w.f.$) is a weighting factor that depends on (E_{Daylight}) and (E_{Limit}) that are the horizontal illuminance at a given point due to daylight and the illuminance limit value, respectively. It is the first metric to account for dynamic temporal occurrences of daylight due to changing climate conditions. Still, DA does not consider target illuminances neither below nor above that threshold, lacking the indications of higher lighting energy demand, or visual discomforts due to sunlight penetration. Rogers and Goldman (2006) proposed CDA by assigning a partial credit to illuminances below a suggested limit. CDA is the percentage of the annual occupied timesteps when the illuminance is over or under a predefined threshold:

$$DA_{\text{con}} = \frac{\sum_i (w f_i t_i)}{\sum_i t_i} \in [0,1], w f_i = \begin{cases} 1 & \text{if } E_{\text{Daylight}} \geq E_{\text{Limit}} \\ \frac{E_{\text{Daylight}}}{E_{\text{Limit}}} & \text{if } E_{\text{Daylight}} < E_{\text{Limit}} \end{cases} \quad (3)$$

Similarly, UDI was developed by Nabil and Mardaljevic (2006), which is the first metric to address different ranges of daylighting performance by introducing the idea of ranged thresholds. UDI is the percentage of the annual occupied timesteps when the illuminance is useful (within the lower and upper limits), underlit ($<$ lower limit), or overlit ($>$ upper limit):

$$UDI = \frac{\sum_i (wf_i \cdot t_i)}{\sum_i t_i} \in [0,1],$$

$$wf_i = \begin{cases} UDI_{Underlit} & \text{with } wf_i = \begin{cases} 1 & \text{if } E_{Daylight} > E_{Upper Limit} \\ 0 & \text{if } E_{Daylight} \leq E_{Upper Limit} \end{cases} \\ UDI_{Useful} & \text{with } wf_i = \begin{cases} 1 & \text{if } E_{Lower Limit} \leq E_{Daylight} \leq E_{Upper Limit} \\ 0 & \text{if } E_{Daylight} < E_{Lower Limit} \vee E_{Daylight} > E_{Upper Limit} \end{cases} \\ UDI_{Overlit} & \text{with } wf_i = \begin{cases} 1 & \text{if } E_{Daylight} < E_{Upper Limit} \\ 0 & \text{if } E_{Daylight} \geq E_{Upper Limit} \end{cases} \end{cases} \quad (4)$$

The approved method of daylight metrics [IES LM-83-12 \(2012\)](#), issued by IES in 2012, then expanded those temporal metrics with $sDA_{300/50\%}$ and $ASE_{1000/250h}$ to account for spatial consideration. $sDA_{300/50\%}$ and $ASE_{1000/250h}$ describe how much percentage of a space receives adequate amount of daylight, and excessive amount of sunlight, respectively. $sDA_{300/50\%}$ is the percentage of a space that receives a minimum target illuminance of 300 lx for temporal fraction threshold of 50% of the annual occupied hours:

$$sDA_{300/50\%} = \frac{\sum_i S(i)}{\sum_i p_i} \in [0,1], S(i) = \begin{cases} 1 & \text{if } s_i \geq \tau t_y \\ 0 & \text{if } s_i < \tau t_y \end{cases} \quad (5)$$

$S(i)$ represents a function, where s_i is the occurrences exceeding the illuminance target at points (p_i), τ is the temporal fraction threshold and t_y is the annual occupied timesteps. $ASE_{1000/250h}$ is the percentage of a space that receives a minimum direct sunlight of 1000 lx for absolute hour threshold of 250 hours of the annual occupied hours:

$$ASE_{1000/250h} = \frac{\sum_i A(i)}{\sum_i p_i} \in [0,1], A(i) = \begin{cases} 1 & \text{if } a_i \geq T_y \\ 0 & \text{if } a_i < T_y \end{cases} \quad (6)$$

where $A(i)$ represents a function, where a_i is the occurrences exceeding the illuminance target at points (p_i) and T_y is the annual absolute hour threshold.

About glare indices, [Einhorn \(1979\)](#) introduced Glare Index, which that was later recommended by CIE, referred to as CGI. It considers direct and diffuse light evaluated on a horizontal plane passing through a viewpoint. Then, [Wienold and Christoffersen \(2006\)](#) modified the original glare formula by presenting DGP, which combines vertical eye illuminance with glare source luminance assessed at the same point. Like most glare indices, DGP requires the source luminance, size and relative position, as follows:

$$DGP = 5.87 \times 10^{-5} \cdot E_v + 9.18 \times 10^{-2} \cdot \log \left(1 + \sum_{j=1}^n \frac{L_{s,j}^2 \cdot \omega_{s,j}}{E_v^{1.87} \cdot P_j^2} \right) + 0.16 \quad (7)$$

where (L_s) is luminance of the glare source luminance, (L_b) is the luminance of the background or field of view from the observer's position, (ω_s) is the angular size of the glare source from the observer's position, (P) is the relative position of the glare source regarding the observer's focal point, and exponents (e), (f) and (g) are weight factors that vary according to glare formulas.

5.3. Data Sources, Sizes and Temporal Granularities

5.3.1. Data Sources and Sizes

The used Data for training and testing machine learning models can be categorized into Data Sources and Sizes, along with Temporal Granularities. Considering the data sources, 2 types can be found in previous studies (Simulation-Derived and Field Measurements). (Fig. 5-i) summarizes their distribution, indicating that 73% of the studies focused on simulation-derived data, while the remaining collected data from field measurements, including HDRI (Inanici, 2013) and experimental survey (Yacine et al., 2017). The Simulation-derived data is used to construct machine learning models, considering both existing spaces and conceptual designs, using simulation tools, such as DIVA-for-Rhino, Radiance, DAYSIM, Ecotect, Honeybee, Groundhog, OpenStudio, EnergyPlus, DELight and ESP-r. (Fig. 5-j) reveals their distribution, where they were used in previous investigational attempts with frequencies of 19%, 15%, 15%, 12%, 12%, 9%, 6%, 6%, 3% and 3%, respectively. The reader can refer to (Ayoub, 2020, 2019b) for additional information on daylight simulation tools capabilities and the used light transport algorithms. Regarding the data sizes, they can be classified into 4 types, according to data sources and their usage (Simulation-Derived: Training and Testing Data, in addition to Field Measurements: Training and Testing Data). Overall, the ranges of data size for Simulation-Derived Training Data are from 5,400 to 195, Simulation-Derived Testing Data are from 729 to 21, Field Measurements Training Data are from 906,540 to 50, and Field Measurements Testing Data are from 140,218 to 31, as shown in (Table 1). (Fig. 10) plots the reviewed studies against data sizes, distributed over a logarithmic scale. It can be concluded that the average ratio between training and testing data is approximately 3:1.

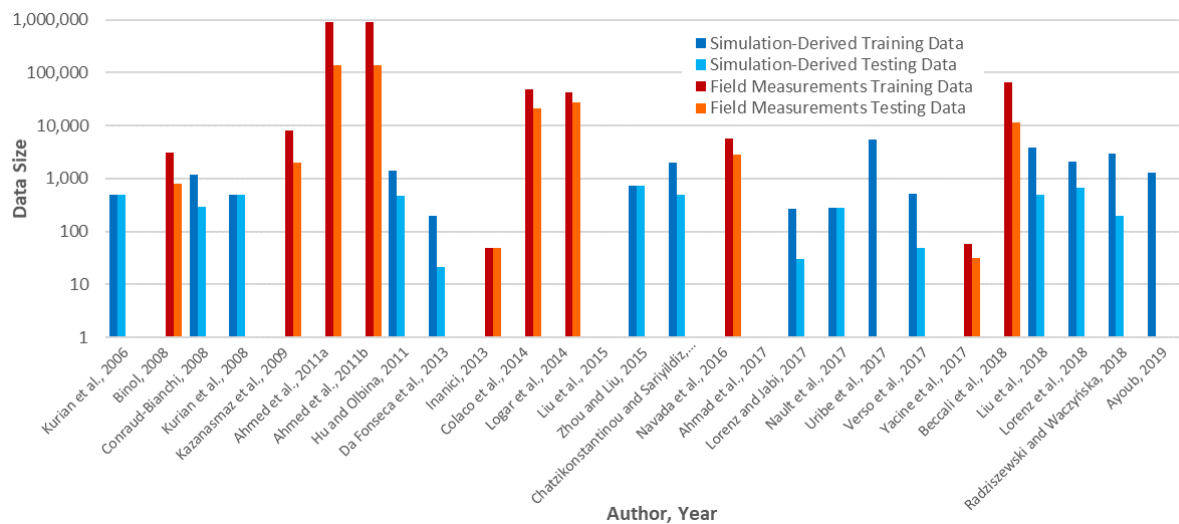


Fig. 10. Distribution of data sizes used to train and test MLAs in previous studies according to author and publishing year.

5.3.2. Temporal Granularities

Temporal granularity is the time of the year and its resolution to consider for a building performance indicator, either a metric or a set of absolute values. In the reviewed studies, they are categorized into 2 main types (Annual and Selected Instances); including 3 subcategorical resolutions (Hourly, Sub-Hourly and Timeseries). (Fig. 5-k) summarizes their overall distribution, showing that almost 43%, 19%, 11%, 9%, 8% and 6% of the studies accounted for Annual (Hourly Basis), Selected Instances (Sub-

Hourly Basis), Selected Instances (Hourly Basis), Annual Timeseries (Hourly Basis), Annual (Sub-Hourly Basis) and Selected Timeseries (Sub-Hourly Basis), respectively; but the remaining studies did not identify the selected temporal granularity. 57% of the reviewed attempts considered Annual predictions, while 38% of them focused on Selected Instances. Considering the temporal resolutions, Hourly resolutions were considered 62% of the attempts, whereas Sub-Hourly were accounted for 34%.

5.4. Evaluation Metrics

Evaluation metrics are accuracy measures used to assess how well machine learning models' training and testing would perform; and this can be achieved by comparing the differences between the predicted and the observed/simulated values. 12 evaluation metrics are identified in the previously reviewed studies, including Root Mean Square Error (RMSE), Percentage of Error (PE), Coefficient of Determination (R^2), Mean Square Error (MSE), Predictive Confidence (PC), Mean Biased Error (MBE), Relative Error Rate (RER), Average Relative Error (ARE), Coefficient of Correlation (R), Coefficient of Variation (CV), Mean Absolute Deviation (MAD) and Mean Absolute Error (MAE). Some of the famous metrics can be acquired from equations (8) to (14).

$$PE = \frac{(\hat{y}_i - y_i)}{y_i} 100\% \quad (8)$$

$$MBE = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i) \quad (9)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i| \quad (10)$$

$$MSE = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2 \quad (11)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2} \quad (12)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (\hat{y}_i - y_i)^2}{\sum_{i=1}^N (y_i - \bar{y}_i)^2} \quad (13)$$

$$CV = \frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2}}{\bar{y}_i} 100\% \quad (14)$$

where $\hat{y}_i$ is the predicted illuminance value for times i ; y_i is the observed/simulated illuminance value for times i ; $\bar{y}_i$ is the average illuminance value; and N is the number of data points used for evaluation. (Fig. 5-I) summarizes their distribution, indicating that 23%, 19%, 16% and 12% of the reviewed studies used RMSE, PE, R^2 and MSE. About 14% of them utilized PC, MBE and RER with the same frequency rate of nearly 5%; while almost 12% used ARE, CV, MAD, MAE and R with the same frequency of about 2%. Other researchers, including (Liu et al., 2015; Uribe et al., 2017), did not specify the used evaluation metrics.

6. Discussion

Next subsections include a discussion of the review results, organized in accordance with the same structural order of the previous section: Scope of Prediction; MLAs; Data Source, Size and Temporal Granularities; and Evaluation Metrics.

6.1. Scope of Prediction

6.1.1. Building Types

In daylighting studies, selecting a specific building type is crucial to determine the applied occupancy schedules and the targeted illuminance levels. Occupancy schedules normally define the hours of the year when a space is occupied by users; hours outside such ranges are not considered in the evaluation of daylighting performance. LEED v4 (USGBC, 2013) and LM-83 (IES, 2012) specify typical occupancy schedules from 8:00 AM to 6:00 PM, including all weekdays. This represents common working hours from 9:00 AM to 5:00 PM, in addition to one hour for arrival and another for departure. Targeted daylight levels vary according to building types and functions in a way that significantly affects the simulation outputs. For suggestions on adequate illuminance levels, the readers are referred the handbooks of the Illuminating Engineering Society of North America (IESNA) (IESNA, 2018) and the American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) (ASHRAE, 2007).

While residential buildings represent a large part of building energy consumption in many countries around the world (Pérez-Lombard et al., 2008), recent reviews, such as (Wang and Srinivasan, 2017) and (Amasyali and El-Gohary, 2018), confirmed that more than 80% of the reviewed researches focused the investigations extensively on non-residential buildings to predict energy consumption by MLAs. This coincides with a much higher percentage, 98%, of the investigated studies in this review that focused on predicting the luminous conditions of non-residential buildings. Such tendency may be attributed to several reasons. (i) Due to the issue of privacy (Jetcheva et al., 2014), it is not easy to regularly collect building performance data from residential buildings. Whereas, the relatively accessible public spaces, such as offices and educational buildings, encourage researchers to investigate those types of buildings instead. (ii) Many test models actually resemble reference office spaces (Reinhart et al., 2013), to achieve some sort of standardized evaluations, allowing to facilitate comparing results from different studies. Still, it is worth mentioning that during the early stages of design, commonly used 3D ‘shoe-box’ test models (Reinhart et al., 2013) that represent standalone buildings do not always reflect reality. In fact, it is more accurate to explicitly account for the actual space geometry (Ayoub, 2019b), surrounding obstructions (Ayoub, 2019a) and external ground, since they have a great impact on daylighting performance (Ayoub, 2019b; Ng, 2001; Reinhart and Walkenhorst, 2001). Nonetheless, such simple models may still be satisfactory for initial estimations using MLAs to achieve generalization in such conceptual stage of design. (iii) Specifically, energy-based predictions using machine learning, as shown in previous reviews, depend largely on a derived data from field measurements and historical datasets. However, this is rarely found in daylighting studies, where many researchers depend mostly on simulation-derived data rather than scarce measurements. (iv) Perhaps the focus on non-residential buildings might stem from researchers who seek to simplify data acquisition by minimizing the time

required for annual simulations; since offices and educational spaces have limited and pre-determined occupancy schedules, while residential buildings are regularly occupied, requiring to perform annual simulations on an hourly basis for all daylight hours; adding more stress to the computations. The more hours of the day are considered, the more stress is added to the computational power, especially when the target is to obtain accelerated preliminary approximations for different design solutions.

6.1.2. Climate Zones and Locations

The inequitable distribution of cities and climate zones in which the studies reflected (Fig. 6) reveals some opposing concerns. For instance, in cooling-dominant climates, such as Hot-Desert, solar heat gain due to excessive sunlight is inevitable, and has long been regarded as a major driver for energy-efficient building envelopes (Ayoub, 2019a, 2018; Elnokaly et al., 2019). The need to decrease heat gains, by minimizing window-to-wall ratio, contradicts with the necessity to maintain adequate level of daylight, by maximizing such ratio (Ayoub, 2018; Xue et al., 2016). Moreover, the precise identification of the study location is of great importance to select adequate weather dataset for building performance simulations. Typically, weather files are metrological data, collected from weather stations, and include single-years that represent historical records of specific locations, for different temporal resolutions, over extended observation periods (Herrera et al., 2017; Moazami et al., 2019). Nonetheless, most of the reviewed studies did not explicitly identify the used weather file, but rather stated the location; and this may entail some confusions. (i) Form which the weather files are created, it is typical to select the nearest weather station to the project location, although there might not necessarily be a nearby station. In some locations, such as California that has 16 different climate zones (EnergyPlus, 2019), few kilometres can make a difference (Reinhart, 2018). Thus, understanding the local weather conditions is needed to select suitable files. (ii) Also, selecting files of unobstructed weather stations, typically in airports, may be unsuitable for daylighting studies in heavily obstructed context, where many weather elements, such as solar irradiances and wind velocities, can significantly be affected by such settings (Ayoub, 2019b). However, studies that rely on simulation-derived data can adjust the used weather files by additional field measurements (Charles and Crawley, 2011), expensive satellite data, or by machine learning to generate synthetic data (Chauhan and Thakur, 2014).

6.2. Machine Learning Algorithms

6.2.1. Problem Types

The majority of reviewed studies tend to investigate regression problems, and this is quite justifiable, as the non-linear nature of internal daylight patterns (Lorenz and Jabi, 2017; Suykens et al., 2012) can tolerably be handled by regression. This also corresponds with the intent to obtain numerical estimations of illuminance values or performance metrics, rather than 2-dimensionally mapping the spatial distribution of internal daylight at sensor points (Carlucci et al., 2015; Galatioto and Beccali, 2016). Classification and clustering problems were only addressed in (Ahmed et al., 2011a; Liu et al., 2015; Zhou and Liu, 2015).

6.2.2. Selected Machine Learning Algorithms and Hyperparameters

As mentioned earlier, after 2012 many MLAs were used to solve regression, classification and clustering problems (Fig. 7). From a daylight simulation perspective, 2012 checkpoint is particularly important to the building design community, as it corresponds with the emergence of multi-core CPUs that were derived by economic and thermodynamic limits (Boix, 2015; Koomey et al., 2011; Parkhurst et al., 2006), at the expense of obsolete single-core designs. Such boost in the computational power was accompanied by the development of: (i) test cases that are used to validate daylight simulation tools (Aizlewood, 1993; CIE, 2006; Mardaljevic, 2001; Schregle and Wienold, 2004); (ii) weather datasets that are representative of extreme climate conditions (CIBSE, 2019; Ferrari and Lee, 2008); (iii) 3- and 5-phase CBDM that brought the accurate and fast modelling of the optical properties of Complex Fenestration Systems (CFS) via Radiance's bi-directional raytracing (Lee et al., 2018; Saxena et al., 2010); (iv) various dynamic metrics that report temporal daylight occurrences (Nabil and Mardaljevic, 2006; Rogers and Goldman, 2006); in addition to (v) a new interesting approach to estimate illuminance from an HDRI (Gardner et al., 2017; Zhang and Lalonde, 2017) and 3D objects (Weber et al., 2018). Not only such factors are bringing innovative daylighting studies, but also can offer new opportunities for potential applications of machine learning in the field of building performance simulation.

Compared to engineering methods, machine learning does not require any prior knowledge about the building physical attributes, including geometry, surrounding context, opening locations and configurations, furniture and material properties. Such information is already embedded in the parameters of the developed model, which in turn, are reflected in the prediction outputs. In this sense, the training and testing datasets are derived from the same feature space and represent the same distribution. To make accurate predictions using MLA, training relevant data is required to construct mathematically-fit models; but they are only applicable in the data range from which they were developed. Once a model is successfully constructed, it can be easily used for accelerated predictions. However, during the early stages of design process, preliminary building designs are iteratively changed to search for optimum alternatives; consequently, the data distribution is modified as well. This can significantly affect the outputs, where it becomes difficult to generalize the daylighting performance for new design proposals. In such case, machine learning models need to be entirely reconstructed and retrained by newly collected set of data. To reduce the effort of such time-consuming work, Transfer Learning between the old and new data domains offer a solution for this complication (Pan and Yang, 2009). Another discrepancy that is associated with using MLAs themselves as holistic approach of univariate model output is that they can be tedious, where every building performance output would entail creating a dedicated algorithm. This is apparent in the reviewed studies and the larger number of investigational attempts they involve.

The choice of suitable MLAs depends on the problem that is being addressed (Wang and Srinivasan, 2017), the collected data type, and model-specific criteria that include accuracy, complexity and speed. However, some decisions that were made in previous studies to use a single algorithm, or compare a number of algorithms, were not supported neither by technical aspects, nor by heuristic methods. The

latter method, while important, is time-consuming, since it requires an extensive empirical trial and error (Lokuciejewski et al., 2010), coupled with a higher degree of experience to be handled properly. Still, machine learning is not commonly practiced by architects (Khean et al., 2018); but with more focused implementations of machine learning in the field, such lack of integration will eventually diminish.

6.2.3. Input Parameters

Apart from (Chatzikonstantinou and Sariyildiz, 2016; Radziszewski and Waczyńska, 2018; Zhou and Liu, 2015), previous studies that predicted annual metrics did not use any Temporal Settings to construct their machine learning models, as such annual data is already embedded in the parameters of the developed models. This coincides with their selected temporal granularity of Annual (Hourly and Sub-Hourly Basis), except for (Yacine et al., 2017), where the authors sought to create models to study the association between lighting quality and glare indices, only for selected timesteps. This is because annual glare studies require creating a hemispherical fisheye image at each timestep, yielding thousands of images per year, putting additional stress on the computations, especially for multiple viewpoints (Ayoub, 2019b). As stated earlier, although the External Obstructions are essential inputs to consider, as they have a profound impact on the internal daylighting conditions (Ayoub, 2019b; Ng, 2001; Reinhart and Walkenhorst, 2001), they were not accounted for that much in previous studies. This can be attributed to the aim to achieve generalization, where simple 'shoe-box' models are sufficient for such initial estimations in the early stages of design.

On the other hand, some of the reviewed studies addressed a small number of input parameters, such as (Colaco et al., 2014; Conraud-Bianchi, 2008; da Fonseca et al., 2013; Kurian et al., 2008; Logar et al., 2014; Yacine et al., 2017). This suggests that they represent isolated cases of which the developed machine learning models are only applicable on, and do not offer generalization. Yet again, even with larger amount of input parameters, many studies did not test the predictive power of the selected inputs. In fact, machine learning models can be constructed from any number of input parameters. However, this does not mean to add as much inputs as possible to the model, since more inputs, while can offer more accuracy, can come on the expense of interpretability. In contrast, less inputs, while can ensure interpretability, but might yield less accuracy and increase the risk of overfitting. Thus, Feature Selection (Kira and Rendell, 1992) can be used to identify those that would contribute to the model construction, while increasing its prediction accuracy and interpretability by discarding insignificant inputs. Evaluation metrics are also used to gauge different combinations of inputs. For clarity, the term (input parameters) refers to the original input data, but the term (feature) denotes those that are identified from the feature selection. When the number of parameters is small, searching all possible combinations of features via brute-force can be used to find the best set. But with higher dimensional datasets, it might be inefficient and time-consuming to go through an even larger number of inputs combinations; especially this would require repeating the model training with every new combination. Thus, other feature selection methods can be used, including wrappers, filters and embedded approaches (Chandrashekar and Sahin, 2014; Vergara and Estévez, 2014). In this sense, feature selection explores the relationship among inputs rather than their value ranges. On the contrary, Sensitivity Analysis (Davis, 1989; Saltelli et al., 2004)

investigates the relationship between inputs, together with a model's architecture, and outputs. In other words, it measures the variation in outputs due to uncertainty in various ranges of inputs and model architectures (Kazanasmaz et al., 2009). Sensitivity analysis can explore the variation in the outputs when inputs vary around their original values, or can account for a wider range of inputs values. Using both of those tests can help designers to (i) realize of the relationship between input and output parameters; (ii) identify the most influential parameters on the predicted outputs; (iii) detect sources of prediction errors that may stem from unexpected relationships between inputs and outputs; and (iv) offer further simplification and interpretability to the developed models.

6.2.4. Output Parameters

Apart from DF, the oldest daylight metrics, researchers did not actually start predicting performance metrics until 2015, as shown in (Fig. 9). This is rather surprising, since the performance metrics that considered temporal occurrences of horizontal illuminances were basically originated back to 1989. This can be attributed to the desire to acquire absolute illuminance values before 2015, rather than performance metrics; especially in studies that considered temporal granularity of Annual Timeseries, such as (Kurian et al., 2008, 2006), as well as the ones that accounted for Selected Instances that include (Ahmed et al., 2011a, 2011b; Binol, 2008; Colaco et al., 2014; Hu and Olbina, 2011; Inanici, 2013; Kazanasmaz et al., 2009; Logar et al., 2014). The majority of the latter cases depended on Field Measurements to collect the training data, as will be explained in the following subsections. Between 2008 and 2015, there was no research on MLAs that focused on predicting daylighting performance metrics, and this can be attributed to two distinct stages of significant developments: before and during that period. (i) Before 2008, during the early years of the new millennium, the first stage has witnessed the progress of multi-core CPUs, coupled with the emergence of validation datasets and test cases (Aizlewood, 1993; CIE, 2006; Mardaljevic, 2001; Schregle and Wienold, 2004), Standard General Sky (CIE, 2003; ISO, 2004), weather datasets of extreme climate conditions (CIBSE, 2019; Ferrari and Lee, 2008), CBDM (Mardaljevic, 2000b; Reinhart and Herkel, 2000) and dynamic metrics that report temporal daylight occurrences (Nabil and Mardaljevic, 2006; Rogers and Goldman, 2006). (ii) During the second stage that started by 2010, the growth of Graphics Processing Units (GPUs) and parallel computation offered unprecedented real-time prediction of building performance (Jones and Reinhart, 2015; McNeil and Lee, 2012; Schardl, 2016). This was coupled by the development of more accurate weather datasets, including International Weather for Energy Calculations (IWECC2) (Joe et al., 2014) and Typical Meteorological Year (TMY3) (Charles and Crawley, 2011), 3- and 5-PM of CBDM (Lee et al., 2018; Saxena et al., 2010) and daylight metrics that consider the spatiotemporal evaluations of daylighting performance (IES, 2012). Thus, it took the research community a few years employ those accelerating developments and MLAs to predict daylighting performance metrics.

Several daylight simulation tools are now supporting a wide range of daylight metrics to report different daylighting performance of various purposes and target values. They capture the spatial and temporal illuminance conditions and encapsulate such information into single values. This seems convenient, particularly in cases that involve annual predictions with thousands of hourly, or sub-hourly, output data.

Still, relying on daylight metrics as outputs for machine learning models yields some inconsistencies. Some of the used daylight metrics in previous studies confusingly do not specify exact threshold values. For example, the definition of DA does not state a specific threshold, but it was suggested to set it at 500 lx (Olbina and Beliveau, 2009). Also, there is no consensus on the illuminance limits for UDI, though many researchers suggested different upper and lower thresholds (David et al., 2011; Mardaljevic et al., 2009; Nabil and Mardaljevic, 2005; Olbina and Beliveau, 2009). The choice of certain thresholds has a significant impact on how the resulting metric value would be. For instance, DA reports the percentage of the annual occupied timesteps when the internal illuminance exceeds a predefined threshold. But if this threshold is changed, the resulting DA value changes as well, given that the internal daylighting condition remains the same in both cases; and this can yield misleading results. Thus, such metrics are not consistent with each other, and do not necessarily yield common resolutions, requiring data post-processing to retrieve the original illuminance values. Alternatively, absolute illuminance values can be more specific in machine learning field of application, especially in cases that require regression; reflecting accurate representation of the study space, rather than indicating the daylighting performance via metrics that rise above or fall below given thresholds. In other words, the process of assessing daylighting by performance metrics should be after predicting absolute illuminance values. In this, (Ayoub, 2018) suggested an interesting approach to convert absolute illuminances into their equivalent metrics, and vice versa.

6.3. Data Source, Size and Temporal Granularities

6.3.1. Data Sources and Sizes

Field measurements are real data collected from sensors, HDRI or surveys. In building performance simulation, they are more accurate than scale models, which may introduce uncertainties related to geometries and photometric properties of materials (Maamari and Fontoynt, 2003; Thanachareonkit and Scartezini, 2010). Therefore, the latter method of data collection was not used in previous studies. Field measurements typically take a long time to collect, and they become inefficient to investigate long-term analysis under a variety of sky conditions. As a result, this data collection approach was only used in studies that accounted for selected timesteps under limited sky conditions. However, such measurements can be useful to examine post-occupancy performance of existing buildings after construction, not during the conceptual design. On the other hand, long-term assessments of internal daylighting using simulation approach would require inputting a complex set of parameters, as indicated before. But they can easily produce estimations of extended periods of time that cover an entire year, under different sky conditions (Mardaljevic, 2000b). Several studies have confirmed that dynamic daylight simulations using DC method, coupled with backward raytracing (Ward, 1994; Ward and Shakespeare, 1998) and Perez All-Weather (Perez et al., 1993) can predict internal illuminance with errors below 20-25%, compared to the limited field observations (Mardaljevic, 2000a; Reinhart and Andersen, 2006; Reinhart and Breton, 2009; Reinhart and Walkenhorst, 2001). This error may seem high, but the sensitivity of human eye can hardly perceive changes in the internal illuminance that vary in a wide range of magnitudes over time (Reinhart, 2011).

2 of the used 10 simulation tools are in fact legacy simulation engines (Radiance and EnergyPlus) that are not easy to use due to their complicated settings and features (Ayoub, 2019b). This, while can be valued by researchers and skilled practitioners, hinders such tools from being casually utilized by unexperienced users (Reinhart and Fitz, 2006). Therefore, many of the newly developed tools are actually visual interfaces to those engines, without integrating some of their complex features. 7 of the used tools are Radiance-based (DIVA-for-Rhino, DAYSIM, Ecotect, Honeybee, Groundhog, ESP-r and OpenStudio). 4 of them include EnergyPlus engine (Honeybee, ESP-r, OpenStudio and DElight); and this allows to conduct daylighting and energy simulations, simultaneously. Radiance is among the oldest simulation tools, which is highly regarded as a standard tool by the building design community (Ayoub, 2019b). For practical applications, both Radiance and its modified version, DAYSIM, have undergone heavy validations against measurements of real room and physical model (Jarvis and Donn, 1997), clear glazing and shading systems (Reinhart and Walkenhorst, 2001), internal illuminance (Reinhart et al., 2006), translucent panels (Reinhart and Andersen, 2006), measurements of goniophotometer (Schregle and Wienold, 2004), and 3-phase method to model CFS (McNeil and Lee, 2012). Radiance was also validated against test cases from the datasets of BRE-IDMB (Mardaljevic, 2000b, 1997, 1995) and CIE (Donn et al., 2007; Geisler-Moroder and Dür, 2008; Osborne, 2012).

While limited data sizes can seemingly be acquired rapidly, they only capture samples of minor effects on the investigated building performance. Therefore, machine learning models should reliably be built using sufficient amount of data; although this would require additional computational effort. Such datasets should also be representative, offering data distribution that is embedded with as much information as possible in the feature space for accurate predictions. It might be surprising that the average data sizes of field measurements that focused on illuminance values are much larger than those of simulation-derived that considered daylight metrics. The latter approach, as explained earlier, captures long-term illuminance conditions and encapsulates thousands of hourly, or sub-hourly, data entries into single values that reflect the building performance for an extended period. However, the uncertainty and non-linearity in simulation-derived calculations are unavoidable (Tregenza, 2017), due to the changing sky luminance distribution patterns that cannot be predicted with complete confidence. Daylighting studies should, still, rely on long-term analysis, which is imperative for inclusive decision-making. Thus, after 2015 (Fig. 10), almost all previous studies depended on daylight simulations to collect the required data, except for (Beccali et al., 2018; Navada et al., 2016; Yacine et al., 2017).

6.3.2. Temporal Granularities

It can be concluded from (Table 1) that Annual predictions were basically related to estimating different daylight metrics using simulation-derived data of reduced sizes; apart from (Ahmad et al., 2017; Kurian et al., 2008, 2006; Liu et al., 2018) that predicted illuminance values. On the other hand, predictions made from Selected Instances were mostly associated with illuminance values, collected by field measurements of larger data sizes; apart from (Yacine et al., 2017) that predicted glare indices of DGP and CGI. Nonetheless, commonly used temporal granularities in previous studies primarily focus on the long-term analysis of occupied hours and daylit hours. The former is related to studies that consider occupancy schedules as a time basis, such as LEED v4 (USGBC, 2013) and LM-83 (IES, 2012); while

the latter is associated with the study of always-occupied spaces, or when the space function changes from time to time, causing alterations to the occupancy schedule.

For whole building performance investigations, several daylight and energy simulation tools, such as DIVA-for-Rhino (Jakubiec and Reinhart, 2011) and Honeybee (Roudsari and Pak, 2014), generate occupancy schedules, where illuminance values can be linked directly to lighting energy consumption for the same temporal granularities, in addition to heating and cooling energies, to estimate electric energy reduction due to natural daylighting. However, this may cause an overestimated energy consumption, if the daylit hours exceed the actual occupancy schedule, in office spaces for instance. Therefore, more realistic occupancy schedules should be used instead for such studies. Another important issue related to selecting temporal granularities is the consideration of weather data. With regard to the same location, the impact of using weather files of different temporal resolutions and periods on building performance was previously investigated, confirming that reducing time intervals from hours into minutes should be considered (Walkenhorst et al., 2002). Also, weather fluctuations cause a significant impact as well, as short-term investigations showed unreliable results and minor effects on daylighting performance (Bellia et al., 2015a, 2015b; Crawley, 1998; Crawley and Lawrie, 2015; Iversen et al., 2013; Yang et al., 2008).

6.4. Evaluation Metrics

After constructing a machine learning model, evaluation metrics are used to assess its performance during training and testing. Evaluating models is an essential step to test their accuracy before predictions can be undertaken in actual applications. Most of the reviewed studies that addressed regression problems depended on RMSE and MSE. Both metrics emphasize on outliers and extreme error values by squaring the differences between the predicted and observed values before averaging them. Since the square term in RMSE and MSE, along with the absolute term in MAE, cancel negative errors, they do not indicate whether the model over- or underestimates the predictions. Alternatively, other metrics, such as PE and MBE, indicate both positive and negative errors.

Selecting an adequate error metric depends mainly on the way the resulting outliers and extreme errors will be treated. Every metric reveals different qualities about the predicted errors. For example, metrics, such as MBE and MAE, are relatively easy to be interpreted, but they do not make error values contribute as much to the overall model accuracy. Whereas the contribution of the square term in MSE makes low errors quadratically bigger than that of RMSE, penalizing the model for making erroneous predictions that slightly differ from their corresponding observations. Thus, the difference between MSE and MAE suggests a small error of large penalty or a larger error of moderate penalty (Twomey and Smith, 1995). However, as MSE values get larger, it becomes harder to understand how well the model performs; thus, RMSE is rather used in such situation, as it converts the squared error values into their initial unit, making them easier to be interpreted and compared with the original observations. In this sense, RMSE, as it aggregates the magnitudes of the errors in a single value, it is highly influenced by outliers; thus, they need to be removed from the dataset before using such metric.

7. Recommendations and Future Research Trends

Based on the reviewed studies, and in attempt to point towards knowledge gaps and missing research opportunities, the summary of recommendations and future trends in the related domain includes:

- **Scope of Prediction:**

- It is recommended to pay more attention towards regularly occupied spaces, including residential buildings, where using simple 'shoe-box' models as a template is satisfactory for initial MLAs estimations to achieve generalization during the conceptual design stage.
- More focus should be put to exploit classification and clustering, as they can be used in many situations that entail classifying daylighting conditions by UDI and DGP.
- It is suggested to expand the use of MLAs to investigate challenging climate zones and harsh weather conditions, while explicitly identifying the studied location along with the used weather dataset file.
- No study has considered investigating the impact of using MLAs to predict complex daylighting situations of CFS. Such systems, including optical light redirecting systems, mirrored louver systems, prismatic window films and angular screens, can be simulated via 3- and 5-phase analysis (Lee et al., 2018; Saxena et al., 2010). They can play an important role as energy-efficient techniques to optimize façades designs, especially for building in dense contexts and highly obstructed built environment; and predicting those system's performance using MLAs would be useful.
- Again, there is a clear gap between the number of research efforts that has been developed for CBDM and the ones that addresses the applications of advanced types of CBDM (3- and 5-phase); and this needs to be addressed in future investigations.
- With the discovery of specialized photoreceptors in human eye (Pauley, 2004), to the author's best knowledge, no study has examined yet the impact of using MLAs to predict non-visual effects of daylighting (Andersen et al., 2012; Mardaljevic et al., 2014; Webb, 2006), which is a vital element of healthy living, responsible for regulating the internal circadian system.
- As discussed earlier, the use of HDRI is a promising approach to estimate illuminance (Gardner et al., 2017; Zhang and Lalonde, 2017). Although such approach offers accuracy and timesaving, it needs more studies, as it only captures instantaneous conditions, not giving a future perspective of the total daylighting performance.

- **Machine Learning Algorithms:**

- To reduce the effort and consumed time to retrain machine learning models due to changes in design components during the conceptual design phase, Transfer Learning can be exploited, where the originally developed model can be reused as the starting point for additional tasks.
- As explained earlier, machine learning is not commonly practiced by architects, and it is urged to incorporate its concepts and ideas into Architectural education. machine learning can have a significant role to play not only to surrogate building performance simulations, but also to quantify and classify large amounts of data, support form-finding and optimize different building components for better energy-efficiency.
- Feature Selection and Sensitivity Analysis should always be incorporated to determine which set of inputs affects daylighting the most, in addition to examining the effect of adjusting weather files, and which weather elements influence the outcomes. Those might be investigated previously (Wang et al., 2019), yet their impact on MLAs has not yet been investigated.
- It is also recommended to integrate absolute illuminance values when using MLAs to predict internal daylighting performance. Arguably, for comparison and optimization, it is more convenient to use single value to evaluate different design scenarios in the same format.

- **Data Source, Size and Temporal Granularities:**

- It is suggested to use the same temporal granularities, preferably daylight hours in daylighting studies, so that different spaces can be compared, incorporating daily and seasonal changes, different climate zones, besides geographical locations and latitudes.

- Long-term resolutions should always be considered, typically on an annual basis. Typical weather files constructed from most recent 20-30 years of observed data, at least 8 years of observed data (Hubbard et al., 2004), can be representative of future situations of the following decade (Eames et al., 2012). Yet again, harsh climate zones of extreme weather excitations (Kalamees et al., 2012; Pernigotto et al., 2014) buildings may encounter should be integrated into typical weather files (CIBSE, 2009; Jentsch et al., 2013; Mylona, 2012).
- To reduce the uncertainty of daylight calculations that stem from continuously changing sky luminance distribution patterns, the statistical significance of simulation results should explicitly be estimated, and possible changes to the building design or its surrounding context should also probabilistically be predicted (Tregenza, 2017).
- Other growing research trends that are related to the study of energy-efficient daylighting strategies are occupancy behavior and control systems (Beccali et al., 2018; Hu and Olbina, 2011), which have a major impact on improving the prediction performance.

8. Conclusion

Machine learning methods should be exploited more frequently in viable resolutions to surrogate complex daylight simulations. From the author's viewpoint, this requires a mutual consideration of both education and practice standpoints. (i) Computational innovations revolutionize how Architecture is being practiced, as tools of building performance simulation, along with 3D modeling and representation courses, are commonly being introduced to students in many Schools of Architecture. Still, machine learning, statistical methods and optimization techniques unfortunately are not. (ii) Architects are now confronted with the urge to adopt new subdomains of Architecture that machine learning are already contributing to. Still, due to the lack of specialized experience, this shift may raise justified concerns among traditional practitioners to integrate what they consider complex methods into their practices. Besides, some practitioners, who got familiar with machine learning methods, might be unaccustomed to handle their difficulties with ease. Indeed, the amount of required training to master such techniques is considerable. This would require more integration of machine learning as early as possible in Architectural education before expecting an effective increase in their use in practice.

Daylight simulation is an active area of research in architecture that provides designers with accurate quantifications of internal luminous conditions. Nonetheless, such approach is time-consuming and computationally expensive, as it demands inputting a complex set of parameters, making it inherently inflexible, especially during the early stages of design, when preliminary assessment of daylighting is required. The review work of this research highlights an alternative approach to exploit the outputs of such complex computational simulations to construct predictive models that build on MLAs. The use of this approach is granting much attention from researchers over past years due to its ability to act as proxy to simulations. This is coupled with the increasing number of studies that have been published previously. Such growing directions consider exploiting different algorithms for both daylighting predictions as well as on glare evaluations. This research particularly focuses on outlining the scopes of prediction, the used algorithms, data sources and sizes, besides evaluation metrics, providing a detailed discussion on the findings of previous studies. Supported by lower error rates, the accumulations of the previously reviewed studies confirmed the ability of MLAs to offer fast and accurate predictions, when compared to complex simulations. Finally, knowledge gaps are also discussed, revealing future trends to use such approaches in the Architectural practice. However, this research

domain is not mature enough, and more investigations in the related architectural applications are strongly recommended.

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